

From Bed Rest to Mars

A literature-derived dataset and machine-learning framework for predicting lower-limb muscle atrophy in spaceflight analogues

Milad Bahari Qaragoz¹ and [Co-author]²

¹AI framework lead: extraction schema, framework design, modelling and results. [Affiliation]

²Scientific lead: literature search, screening, extraction and physiological interpretation. [Affiliation]

Version	0.1 — initial draft for internal review
Date	21 September 2026
Dataset	dataset_v1.1.csv, git tag dataset-v1.1, SHA-256 dd214c6a. . . b9
Code and results	branch feat/jev-figures, commit 12b254f
Repository	https://github.com/MiladBahariQaragoz/BedrestToMars

Status of this document. This is the first complete draft of the written report. It sets out the full record of what was done and found, and has not yet been condensed. Every table and figure is generated from the frozen dataset and the committed results files; numbers in the text were checked against the same files. Items that still need a decision or a co-author's input are marked [Draft note: like this]. The accepted abstract remains the contract for the presentation; Section 7.7 lists where the results now differ from its wording.

Abstract

Introduction. Musculoskeletal deconditioning is one of the obstacles to long-duration spaceflight, and head-down and horizontal bed rest are its established terrestrial analogues. Three decades of bed-rest studies report lower-limb muscle loss, but they report it campaign by campaign, with different muscles, durations, imaging methods and countermeasures. We set out to turn that literature into one documented dataset and to build a modelling framework that estimates, and honestly validates, how much muscle is lost after a given period of unloading and which muscles lose most.

Methods. A systematic search of PubMed, Scopus, Web of Science and the NASA Technical Reports Server (2013 onwards; 5,731 records), a pre-search corpus of older studies and one campaign from NASA's open bed-rest archive were extracted into a frozen schema in which one row is one study, arm, muscle and measurement occasion. Papers reporting the same participants were mapped to a single campaign, and the campaign, not the paper, is the unit of every validation. The framework has three tiers: (1) a three-level meta-regression, nonlinear in the day of the scan, with random intercepts for campaign, study and muscle and cluster-robust inference; (2) four machine-learning families (ridge regression, random forest, support vector regression and gradient boosting) compared with a duration-only curve under leave-one-campaign-out cross-validation with nested tuning; and (3) a post-hoc probabilistic forecast by a language model (TypeSafe's Jev) that reads each held-out campaign's description alongside the other campaigns' data, checked by ablations, a recognition probe and a validation battery. Every bar was declared before the answer it governs.

Results. The frozen dataset holds 742 rows from 52 studies (1992–2026) and 36 independent campaigns; the modelling subset holds 346 rows from 32 campaigns, scanned between day 5 and day 119. Loss follows a curved path: -2.99 pp per doubling of days (95% CI -3.44 to -2.53), a time constant of 60 days and an eventual loss of -17.3% (-20.9 to -13.7) for the reference scenario, although logarithmic, saturating and spline forms are indistinguishable (within 1.3 AIC points). Which muscle is measured matters most: 30% of the variance lies between muscles within a campaign, plantar flexors lose 5.69 pp more than knee extensors at day 60 ($p = 0.001$) and 5.60 pp more than dorsiflexors ($p = 0.003$), and countermeasure arms lose 3.73 pp less than controls ($p = 0.002$). No machine-learning family beat the duration curve (best: random forest, 3.18 vs. 3.16 pp out-of-campaign MAE), a null result pre-committed to in advance. The language-model forecast was 0.42 pp more accurate than the curve (95% CI 0.13 to 0.73; 20 of 32 campaigns), survived the removal of every study-identifying detail and every presentation check, but missed its declared 15% improvement bar and its ranges were overconfident.

Conclusions. The published bed-rest literature supports a quantified, curved duration–response and a muscle-specific vulnerability led by the plantar flexors. At 32 independent campaigns, flexible tabular learners add nothing to a simple curve; the sample size is the number of campaigns, not the number of rows. The dataset and the framework are the contribution, and both are reproducible from one command.

Keywords: bed rest; head-down tilt; muscle atrophy; spaceflight analogue; meta-regression; leave-one-cohort-out cross-validation; machine learning; probabilistic forecasting.

Contents

1	Introduction	1
1.1	Musculoskeletal deconditioning and long-duration missions	1
1.2	Bed rest as a terrestrial analogue	1
1.3	The gap	1
1.4	Aim and contributions	1
1.5	Structure of the report	2
2	Background	2
2.1	Unloading atrophy is not uniform across muscles	2
2.2	Countermeasures	3
2.3	Pooling aggregate data from clustered studies	3
2.4	Machine learning at small sample sizes	3
2.5	Positioning: what machine learning adds to meta-regression	3
3	Methods I: building the dataset	3
3.1	Design principles	3
3.2	Literature search	4
3.3	Eligibility criteria	5
3.4	Screening	5
3.5	The pre-search corpus	5
3.6	Extraction schema	6
3.7	Source verification and reconciliation	6
3.8	The campaign map	7
3.9	Open NASA data	7
3.10	Build, quality control and freeze	7
3.11	The modelling subset	8
3.12	Target and exposure	8
3.13	Muscle classification	9
4	Methods II: the modelling framework	9
4.1	Overview	9
4.2	Features	9
4.3	Tier 1: the three-level meta-regression	11
4.4	The duration-only baseline	12
4.5	Tier 2: the comparative machine-learning families	12
4.6	Tier 3: a probabilistic language-model forecast	13
4.7	Success criteria declared before modelling	15

4.8	Sensitivity analyses	15
4.9	Software and reproducibility	15
5	Results I: the corpus and the dataset	16
5.1	Search and screening	16
5.2	The dataset	17
5.3	The modelling subsets	18
5.4	A first look at the data	18
6	Results II: the three tiers	19
6.1	Tier 1: the duration–response	20
6.2	Tier 1: muscle-specific vulnerability	21
6.3	Sensitivity analysis S6: does the weighting matter?	22
6.4	The duration-only baseline	23
6.5	Tier 2: the comparative machine-learning families	23
6.6	Tier 3: the probabilistic forecast	24
6.7	Tier 3: is the gain signal or recognition?	25
6.8	Tier 3: does it hold up?	27
6.9	The effect of correcting the time axis	29
6.10	Assessment against the declared criteria	29
7	Discussion	30
7.1	Principal findings	30
7.2	The duration–response	30
7.3	Muscle-specific vulnerability: mechanical role	31
7.4	Why flexible learners did not help	31
7.5	What tier 3 does and does not show	31
7.6	Implications for mission planning	32
7.7	Reconciliation with the accepted abstract	32
8	Limitations	32
9	Status of the work and remaining tasks	34
10	Conclusions	34
	Data and code availability	34
	Acknowledgements	35
	Author contributions	35

Use of AI-based tools	35
Competing interests and funding	35
References	36
A Search strategies	41
B Extraction schema	42
C The 36 campaigns	43
D Muscle classification	45
E Tier-1 coefficients	47
F Tier-3 recognition probe by campaign	48
G Reproducing the results	48

List of Figures

1	The framework	10
2	Flow of records from search to modelling subset	17
3	The 32 campaigns of the modelling subset	19
4	Duration–response: control-arm observations and the tier-1 curve	20
5	Muscle-family ranking at day 60	22
6	Tier 2 against the duration curve	24
7	Tier 3: gain by held-out campaign	25
8	Tier 3: forecast against observed value	26
9	Tier 3: recognition probe	27
10	Tier 3: the gain under every check	28

List of Tables

1	Project phases. The written report and the presentation slides are phase P5.	4
2	Eligibility criteria. A study was included when every row held.	5
3	Source verification of the extraction (4 September 2026).	6
4	The three tiers.	10
5	Success criteria for tiers 1 and 2, declared before modelling (project plan, section 8). . .	15
6	Declared sensitivity analyses and their status on 21 September 2026. ★ = must be shown in the report.	16
7	Characteristics of the frozen dataset	18
8	From the frozen dataset to the modelling subsets	18
9	Unadjusted control-arm means by functional class	19
10	Tier 1: the three duration forms	20
11	Tier 1: the fitted curve at selected days	21
12	Tier 1: variance components	21
13	Tier 1: muscle-family ranking at day 60	21
14	Sensitivity analysis S6	22
15	Tier 2: out-of-campaign error of the baseline and the four families	23
16	Tier 2: stability of feature importance	24
17	Tier 3: forecast accuracy and calibration	24
18	Tier 3: ablations	26
19	Tier 3: validation battery	28
20	The time-axis correction	29
21	Assessment against the declared criteria	30
22	Reconciliation with the accepted abstract	33
23	The extraction schema	42

24	The 36 campaigns of the dataset	43
25	Muscle classification	45
26	Tier-1 fixed effects	47
27	Recognition probe by campaign	48

1 Introduction

1.1 Musculoskeletal deconditioning and long-duration missions

Skeletal muscle adapts to the load it carries. When the load is removed—in weightlessness, or when a healthy person lies in bed for weeks—the muscles that normally hold the body up against gravity lose mass and strength. For a crew on a transit to Mars, which lasts on the order of six months, that loss is not an abstraction: a crew has to be able to stand, move and work on arrival, and countermeasures have to be planned before departure. A planner therefore needs two numbers that the literature does not currently provide in one place: how much of a given muscle is lost after a given number of days of unloading, and how much that depends on anything other than time and the muscle in question.

1.2 Bed rest as a terrestrial analogue

Actual spaceflight data on lower-limb muscle size are scarce, and the few available cohorts are small. The established terrestrial analogue is bed rest: healthy volunteers remain in bed, usually tilted six degrees head-down so that fluid shifts towards the head as it does in orbit, for days to months, while their muscles are scanned before, during and after the unloading. Related models unload one leg (unilateral lower-limb suspension) or the whole body in water (dry immersion). Over three decades these campaigns—run at NASA, MEDES in Toulouse, the Charité in Berlin, the DLR in Cologne, and elsewhere—have produced a large and heterogeneous literature on muscle loss [1–7].

1.3 The gap

Each campaign reports its own muscles, at its own time points, with its own imaging protocol, participants and countermeasures. A reader who wants to know how much soleus is left after sixty days has to read dozens of papers and reconcile them by hand. Two further difficulties make naive pooling misleading. First, many papers report the same participants: a single campaign can yield five or more publications, so counting papers overstates the evidence and, worse, leaks information across the folds of any cross-validation that treats papers as independent. Second, the numbers themselves are heterogeneous: some papers report muscle volume by magnetic resonance imaging (MRI), others cross-sectional area by computed tomography (CT), lean mass by dual-energy X-ray absorptiometry (DXA), or thickness by ultrasound; some report a composite such as the quadriceps together with its four heads; and some report only a percentage change.

1.4 Aim and contributions

The project had one aim, taken from the accepted abstract: to develop a literature-derived dataset and a machine-learning framework for predicting lower-limb muscle atrophy in spaceflight analogues. It makes four contributions.

1. **A frozen, documented dataset.** 742 comparisons against baseline from 52 studies and 36 independent campaigns, each traceable to a table, figure or page of its source, extracted into a schema frozen before extraction began and released under a hash-verified version tag ([Sections 3 and 5](#)).
2. **A campaign map.** Every study is assigned to the campaign that produced its participants, which makes the campaign—not the paper—the unit of validation ([Section 3.8](#)).
3. **A three-tier modelling framework** in which a three-level meta-regression carries the scientific estimate, four machine-learning families test whether flexible models find structure the curve misses, and a probabilistic language-model forecast tests whether a model that reads each campaign's

description can do what the numeric columns cannot ([Section 4](#)).

4. **A declared evaluation.** Every threshold, reading rule and sensitivity analysis was written down before the answer it governs, and every number in this report is regenerated from the frozen dataset by one command ([Section 4.9](#)).

The report is written as a framework-and-dataset contribution, not a performance contribution. With 32 independent campaigns in the modelling subset, no cross-validated error will impress a machine-learning audience, and it does not need to: the questions that matter here are whether the estimates are physiologically plausible, whether the uncertainty is stated honestly, and whether the approach could ever inform mission planning.

1.5 Structure of the report

[Section 2](#) summarises the physiological and methodological background. [Section 3](#) describes how the dataset was built and [Section 4](#) how it was modelled. [Section 5](#) describes the corpus and the dataset, and [Section 6](#) reports the three tiers. [Section 7](#) interprets the findings, states the limitations in full, lists the work that remains and reconciles the results with the accepted abstract. The appendices hold the verbatim search strategies, the extraction schema, the campaign table, the muscle classification and the full model outputs.

2 Background

[Draft note: The physiological background (PLAN.md task 1.7, docs/background.md) is the scientific lead's section and has not been drafted yet. The text below is a placeholder written from the corpus and the design document so that the report reads end to end; it should be replaced or rewritten by the co-author.]

2.1 Unloading atrophy is not uniform across muscles

Bed-rest studies that image individual muscles find that loss is concentrated in the muscles that carry body weight in upright stance. In the first Berlin Bed-Rest Study, MRI of the whole lower limb showed the calf, and the soleus and gastrocnemius in particular, among the most affected muscles, with the vasti losing more than rectus femoris and the hip musculature losing less [3]. The same group showed that atrophy is heterogeneous even within a single muscle, depending on where along its length the slice is taken [4]. The 90-day Long-Term Bed Rest study at MEDES documented large losses in the knee extensors and plantar flexors and their partial prevention by flywheel resistance exercise [2, 8], and a 17-week horizontal bed rest at NASA established the longest unloading duration in the corpus [1]. A systematic review of bed-rest studies described the loss of strength and muscle size as nonuniform across muscle groups and over time [9].

Two explanations are usually offered for this selectivity: fibre-type composition, since the soleus is dominated by slow (type I) fibres, and mechanical role, since the soleus and the vasti resist collapse of the ankle and knee under body weight. The two are confounded in most muscles, but not in all. Tibialis anterior is also slow-fibre dominant [10] yet lifts the foot rather than carrying body weight, and rectus femoris extends the knee but also flexes the hip. How the framework classifies these muscles, and why, is set out in [Section 3.13](#).

2.2 Countermeasures

Most recent campaigns include one or more countermeasure arms: resistance or flywheel exercise, aerobic exercise, reactive jumps, whole-body vibration, artificial gravity by short-arm centrifugation, neuromuscular electrical stimulation, blood-flow restriction and nutritional supplementation, alone or in combination. The corpus contains examples of each [e.g. 6, 11–15]. At the number of campaigns available here, countermeasures can be modelled only coarsely (Section 4.2).

2.3 Pooling aggregate data from clustered studies

The dataset consists of published group means, several per paper and several papers per campaign. The standard statistical treatment of such data is aggregate-data meta-regression with a multilevel random-effects structure: effect sizes nested in studies, studies nested in higher-level clusters [16–18]. Its practical advantage here is that it does not require the within-study covariances between muscles, which no bed-rest paper reports. Dose–response meta-analysis offers flexible, spline-based forms for the exposure term [19]. Two cautions from that literature shape the design: the number of study-level covariates that can be supported is roughly one per ten independent studies [20], and many published meta-regressions fail on exactly the points of clustering, covariate count and ecological inference [21].

2.4 Machine learning at small sample sizes

Flexible learners such as random forests [22], gradient boosting [23] and support vector regression [24] are data hungry: simulation studies suggest that they need many times more events per variable than classical regression before their performance stabilises [25]. When observations are clustered, random cross-validation splits place correlated observations on both sides of the fold boundary and overstate performance; grouped (blocked) cross-validation is the recommended remedy [26]. Model-agnostic importance measures such as SHAP [27] describe what a fitted model relies on, not what causes the outcome, and at small sample sizes their rankings can be unstable across folds.

2.5 Positioning: what machine learning adds to meta-regression

The sharpest methodological question this project faces is why machine learning should be used at all when mixed-effects meta-regression is the established tool. The position taken here is that it should not replace the meta-regression, and does not. The meta-regression (tier 1) is the estimator: it produces coefficients with intervals, and it is the primary result. The machine-learning families (tier 2) are a test: given the same rows and the same leave-one-campaign-out discipline, do flexible models find structure—interactions, thresholds, nonlinearities in covariates—that the curve misses? A null answer is informative, because it says how much signal the published literature contains. The language-model forecast (tier 3) asks a different question again: whether a model that reads the *description* of a campaign, and the other campaigns’ data in context, can extract information the numeric columns do not carry. Its outputs are probability distributions and are scored with proper scoring rules [28, 29].

3 Methods I: building the dataset

3.1 Design principles

Three principles governed every step, and they explain several choices that would otherwise look unusual.

1. **Declare before answering.** The extraction schema was frozen before extraction started; the framework

design, the success thresholds, the sensitivity analyses and every reading rule for the post-hoc analyses were committed to the repository before the first result they govern was produced. Where a rule was added after a result was known, the report says so.

2. **The campaign is the unit.** Two papers that report the same participants are one observation of the underlying physiology, not two. Every validation in the project holds out whole campaigns, and every interval is computed by resampling campaigns.
3. **Nothing is typed twice.** The dataset is frozen under a version tag and a SHA-256 hash; the framework refuses to run against a file that does not match; every number and figure is regenerated from that file by one command ([Section 4.9](#)).

The work was carried out by two leads in parallel tracks that met at a frozen interface: the scientific lead ran the search, screening and physiological interpretation; the AI framework lead designed the schema, the framework and the models. The phases and their dates are summarised in [Table 1](#).

Table 1. Project phases. The written report and the presentation slides are phase P5.

Phase	Dates (2026)	Content and outcome
P0 Kickoff	4 Sep	Repository, branch model, frozen extraction schema, decisions on the pre-search corpus
P1 Literature	4–11 Sep	Search of four databases, title/abstract and full-text screening, extraction, campaign map
P2 Integration	12–14 Sep	Source verification, reconciliation, quality control; dataset_v1.0 frozen (14 Sep)
P3 Design	15–18 Sep	Framework design document, configuration, modules, leakage guard, baseline
P4 Modelling	16–21 Sep	Tier 2 (18 Sep); tier 1 and dataset_v1.1 (19 Sep); time-axis correction, tier 3 with ablations and validation, figures (21 Sep)
P5 Report and slides	22–28 Sep	This document; the presentation deck

3.2 Literature search

Four sources were searched on 4 September 2026: PubMed, Scopus, Web of Science Core Collection and the NASA Technical Reports Server (NTRS). Each query combined two concept blocks, both required: *A*, the unloading model (bed rest, head-down tilt, dry immersion, limb suspension, simulated microgravity, disuse and their variants), and *B*, the muscle outcome (atrophy, muscle volume, mass, size, cross-sectional area, lean mass, thickness, and the names of the principal lower-limb muscles). Imaging-modality and countermeasure blocks were deliberately left out of the first pass so as not to lose papers whose abstracts omit them. Rodent studies were excluded in the query where the syntax allowed. Every search was limited to publications from 2013 onwards; older work enters the dataset only through the pre-search corpus ([Section 3.5](#)). The NTRS search used five short keyword queries, because a technical memorandum is not expected to be indexed by the journal databases. The queries are reproduced verbatim in [Section A](#).

Embase, Cochrane CENTRAL, Europe PMC full-text search, trial registries, citation chasing and hand-searching were not run. Embase was unavailable for lack of institutional access; its absence is a known limitation because it indexes conference abstracts the other databases do not.

Known-item test. Before a query is trusted, it should return the papers already known to be relevant. Because of the 2013 date limit, only two of the ten known modelling papers fell inside the search window; both were retrieved by all three journal databases. The test with the date limit removed was not run.

3.3 Eligibility criteria

The criteria in Table 2 were written before screening began. Two judgement calls were settled in advance: total limb volume by anthropometry or plethysmography is not a muscle outcome, because fluid shifts confound it; and cast immobilisation and step reduction are not unloading models for this dataset, because their mechanical and postural picture differs from bed rest.

Table 2. Eligibility criteria. A study was included when every row held.

Axis	Requirement
Population	Humans, adults (≥ 18 years), any sex, healthy or without a disease that itself causes muscle wasting
Exposure	A ground-based unloading model with a stated duration: horizontal or head-down bed rest, dry immersion, unilateral lower-limb suspension. Spaceflight eligible but flagged separately
Duration	At least 5 days of continuous unloading
Outcome	At least one lower-limb muscle-tissue measure at baseline and at one or more later time points: volume, cross-sectional area, thickness, or lean or muscle mass, by MRI, CT, DXA, pQCT or ultrasound
Reporting	Enough to compute a percentage change per arm: baseline and follow-up values, or a printed change
Type	Primary study; reviews were mined for references and never extracted
Language	English or German; other languages recorded rather than silently dropped
Exclusion codes	not_human, not_unloading_model, duration_too_short, no_muscle_outcome, upper_limb_only, no_baseline_or_followup, disease_causes_wasting, review_or_editorial, duplicate_report, no_full_text, language

3.4 Screening

Records from the four exports were merged and de-duplicated on DOI and normalised title (`framework/merge_search_exports.py`); duplicates were marked rather than deleted. Title and abstract screening ran in two stages: a deterministic triage against the criteria (`framework/triage_screening.py`), which sorted records into exclusions with a reason, a priority set and a *maybe* set, followed by a human reading of the priority set and the top of the *maybe* set. Full texts were retrieved preferentially as machine-readable XML from Europe PMC, which preserves tables, and otherwise as PDF. The remainder of the *maybe* set was not screened in the time available; those records are reported as unscreened, not as excluded.

3.5 The pre-search corpus

Fifteen documents were held by the team before the search began. Each was read in full and classified with a written reason: nine were modelling studies, one a methods paper that re-analyses a cohort already extracted, four were vascular studies retained only as context, and one—a study of boys with Duchenne muscular dystrophy—was excluded because the disease itself causes wasting. The nine modelling studies are the dataset’s only source of data published before 2013 and include its longest unloading duration, 119 days [1]. They are a convenience corpus, not a search result, and are counted on their own line in the flow diagram.

3.6 Extraction schema

The schema (`data/schema.md`) defines one row as *one comparison against baseline*: one study, one arm, one muscle, one measurement occasion, carrying the baseline value, the follow-up value and the percentage change. There are no baseline-only rows. The 61 columns fall into groups for study identity and provenance, unloading design, arm and participants, what was measured, values, and quality control; muscles are named from a controlled vocabulary that can be extended only by a change to the schema. Ten rules accompany the schema; the ones that carry the analysis are:

- **Never impute silently**: a missing value is recorded as missing, and that is itself a fact about the literature.
- **Sign convention**: `pct_change` is negative for atrophy, everywhere.
- **Recompute every percentage** where both values are printed, and compare it with the printed change.
- **Composites are not sums**: `triceps_surae` is not the sum of soleus and gastrocnemius, and a composite and its components must never enter a model together unresolved.
- **One cohort_id per campaign**, not per paper.
- **Recovery is a different question**: a recovery row never enters a duration–response model without an explicit flag.
- **n_analysed, not n_arm, is the weight**, because dropouts are routine.
- **Every row is traceable in under a minute** through `source_file` and `page_ref`.
- **Digitised figures are labelled** as such (`data_source = figure_digitized`).

Twelve validation checks run as a script (`framework/validate_extraction.py`) so that a malformed table fails before it is committed. The schema is summarised in [Section B](#).

Two estimators in one column. Where a paper prints the mean of the participants’ individual percentage changes and it differs from the value recomputed from the group means by more than rounding, the printed value is kept (`pct_of_individual_means`); otherwise the recomputed value is kept (`pct_recomputed_from_group_means`). The two are different estimators when the change correlates with baseline size, and every row records which it carries.

3.7 Source verification and reconciliation

On 4 September 2026 the scientific lead re-extracted every provenance record then present from the original papers, with AI assistance, and compared the result with the extraction tables. The check was staged: three studies (26 rows) to test the method, then 113 rows (all 14 figure-derived rows and 99 table-derived rows), then all 580 records. The outcome is summarised in [Table 3](#).

Table 3. Source verification of the extraction (4 September 2026).

Measure	Result
Provenance records checked	580 (537 main, 29 partial, 14 figure)
Recovered from the source	580 (575 high confidence, 5 medium)
Agree without correction	556
Flagged	24 rows, plus 3 duplicated observations
Unresolved numeric conflicts	0
Percent-change gap: median / mean / maximum	0 / 0.012 / 1.04 pp

The flagged rows were resolved on 14 September and each resolution is logged in `data/reconciliation_`

log.md. Eight rows had been coded as MRI but measured lumbar multifidus by ultrasound and were corrected. Sixteen rows printed a whole-number percentage that differed from the recomputed value; because a whole number disagrees with a precise value only once the gap exceeds its own rounding, the printed value was kept where the gap exceeded 0.5 pp (five rows) and the recomputed value otherwise. Duplicated observations were flagged and counted once, and two corrections were made to the campaign map.

This check is *AI-assisted independent source verification*. It is not a human double extraction, and because the assistant had seen some rows earlier it is not fully blinded; accordingly, `double_extracted` remains FALSE on every row. The 165 rows from the pre-search corpus were added after the check and have not been verified by a second person.

3.8 The campaign map

`data/cohorts.csv` assigns every study to the campaign that produced its participants, using trial registry identifiers where they exist and the text of the papers otherwise. Each assignment carries a note and a verified flag. Examples of the dependence it captures: the 90-day Long-Term Bed Rest study at MEDES is reported by four papers in the dataset [2, 8, 30, 31]; the WISE-2005 women’s campaign by five [5, 30, 32–34]; and the McGill 14-day campaign in older adults by three that share one registry identifier [7, 35, 36]. The map also separates campaigns that an author-based grouping would merge: the two Berlin bed-rest studies, BBR1 (56 days, horizontal) and BBR2-2 (60 days, head-down), are distinct campaigns with distinct participants [3, 4]. The resulting 36 campaigns are listed in [Section C](#).

3.9 Open NASA data

On 19 September 2026, seven campaign folders of derived, individual-participant bed-rest data were downloaded from the NASA Life Sciences Portal (NLSP) as open US government data, with every file’s identifier and SHA-256 recorded in a manifest (`framework/fetch_nasa_nlsp.py`). Only one folder was pooled into the dataset. Four of the seven belong to campaigns the dataset already holds as literature rows—three to the 70-day NASA SPRINT campaign and one to WISE-2005—and pooling them would have counted the same participants twice. Two carried no phase labels, so a baseline would have had to be guessed. The seventh, whole-body DXA from the NASA Flight Analog Project’s UTMB Campaign 3 [37], was pooled for the three blocks whose own scan dates confirm a 90-day, 6° head-down bed rest (13 participants). Leg lean mass was aggregated to five cohort-level rows (`framework/extract_nasa.py`): four occasions during bed rest and one in recovery. These are the only rows in the dataset computed from individual measurements rather than read from a publication. Three of the four in-bed-rest occasions (days 30, 45 and 58) rest on a single participant and are weighted accordingly (`n_analysed = 1`).

3.10 Build, quality control and freeze

`framework/build_dataset.py` stacks the four extraction tables (main, partial, figure-derived and NASA) with a `source_table` column in front of the 61 schema columns, leaves out ten rows that are second copies of an observation another row already carries, and checks the result across tables: unique `row_id`, known campaigns, `pct_change` within range, and the minimum viable dataset declared in the plan (60 rows, 8 campaigns, 4 muscle groups, 4 durations). Nothing else is filtered: recovery, spaceflight, dry immersion, DXA, partial and figure-derived rows are all in the file, each marked by its own column, so that the modelling subset is chosen in the configuration rather than built into the data.

Version 1.0 was frozen and tagged on 14 September 2026 (737 rows, 51 studies, 35 campaigns). Version 1.1, frozen on 19 September, added the five NASA rows and one campaign, and is the version every result

in this report was fitted against. The build reproduces each version byte for byte from its tables and refuses to overwrite a frozen file; the loader verifies the SHA-256 on every run. The one schema change v1.1 required—a repository value for `data_source`, so that a row computed from an open archive can say so—is recorded in the reconciliation log. [\[Draft note: this schema change carries one lead's signature; the frozen schema requires both.\]](#)

3.11 The modelling subset

The modelling subset is defined by three predicates declared in `framework/config.yaml` and applied in order:

1. **During unloading only** (`phase = bed_rest`). Recovery rows answer a different question and are reserved for a reconditioning sensitivity analysis.
2. **Lower-limb muscles only**. Multifidus, lumbar erector spinae, quadratus lumborum, psoas and iliopsoas are dropped: trunk muscles unload differently and would answer another question inside the same coefficient.
3. **One tissue per measurement occasion**. A measurement occasion is a campaign, arm, scan day, modality and outcome. Where a paper reports both a composite (for example the quadriceps) and components that cover it (its four heads), the components are kept and the composite dropped, because a model given both fits the same tissue twice. Where no components are present the composite is kept and flagged. Whole-segment measurements (whole thigh, whole calf, whole lower limb) are never decomposed or suppressed.

After predicates 1 and 2 the subset holds 425 rows from 42 studies and 32 campaigns, 299 from control arms and 126 from countermeasure arms, with 13 planned durations and scans on 24 distinct days between day 5 and day 119. Of these rows, 369 are MRI (314 MRI volume), 29 DXA, 15 CT and 12 ultrasound. Both a composite and its components were present on 20 of the 93 measurement occasions.

Two subsets, for two claims. Seven campaigns report only whole-segment measurements, and six of them sit between 5 and 21 days, which is where the duration curve bends most. Excluding whole-segment rows would cost a fifth of the campaigns and the short-duration evidence with them. Subset A, used for the duration–response and for every tier-2 and tier-3 comparison, therefore keeps them as a `whole_limb` level of the muscle family (346 rows, 32 campaigns); subset B, used for the muscle ranking, keeps named muscles only (304 rows, 25 campaigns). [Table 8](#) in [Section 5](#) walks the dataset down to both.

The sample size is 32, not 346. The largest campaign, MEDES LTBR, contributes 86 rows to subset A and the two Berlin campaigns another 82 between them. Every design decision in the framework follows from treating the number of independent campaigns as the effective sample size.

3.12 Target and exposure

Target. The target is `pct_change`, the percentage change in muscle size from the same arm's baseline, negative for loss. In subset A it ranges from -30.0% to $+24.3\%$ (mean -8.0% , median -6.8%); the positive values are real, coming from countermeasure arms and small muscles under compensatory loading. No transformation is applied, so that coefficients read directly in percentage points. The dispersion reported with each value is heterogeneous: of the 425 lower-limb unloading rows, 290 carry a standard deviation, 88 a standard error, 6 a 95% confidence interval, 2 an interquartile range and 39 nothing; and most papers report the spread of the baseline rather than of the change.

Exposure: the day of the scan. The exposure is the number of days of unloading at the time of the scan (`timepoint_days`), not the campaign’s planned length (`duration_days`). The first runs of tiers 1 and 2 used the planned length, which placed 200 of the 346 rows in subset A—every scan taken before bed rest ended, 120 of them more than a week before—at the end of their campaign, so that a day-14 scan in a 56-day campaign entered the model as 56 days. The error was found and corrected on 21 September 2026; it is a correction of the implementation, since the design already named the scan day as a within-campaign variable, not a new analysis choice. The exposure is now declared once in the configuration and read through one function by every tier. The effect of the correction is reported in [Section 6.9](#).

3.13 Muscle classification

The schema’s `muscle_function_class` was empty in every extracted row, and `composite_of` was populated on only 50 of the 189 composite rows in the modelling subset. Both are therefore supplied by a reviewable table, `data/muscle_map.csv`, joined at load time rather than written into the frozen dataset, so that an assignment can be argued with and re-run without a new dataset version. It gives each of the 51 muscles a family (plantar flexors, dorsiflexors, evertors, knee extensors, knee flexors, hip extensors, flexors, abductors, adductors and rotators, trunk extensors, whole segment), a functional class, its components where it is a composite, a fibre-type profile and a written rationale; the loader refuses to run if the dataset names a muscle the table does not cover. The full table is in [Section D](#).

Function was assigned by mechanical role first and fibre-type composition second [10]: 19 muscles are antigravity extensors, 8 flexors and 24 of mixed function. Three judgement calls are flagged for review. Rectus femoris is *mixed*, not an antigravity extensor, because it extends the knee but flexes the hip and loses less than the monoarticular vasti. Tibialis anterior is a *flexor* despite being slow-fibre dominant, because it lifts the foot rather than carrying body weight; it is the cleanest test in the corpus of role against fibre type. And the deep posterior compartment—flexor digitorum longus, flexor hallucis longus and tibialis posterior—sits with the antigravity extensors, because what these muscles do at the ankle is plantar flexion. [Draft note: the classification awaits the scientific lead’s physiological sign-off, which the plan makes a blocking condition for quoting coefficients by class.]

4 Methods II: the modelling framework

4.1 Overview

The framework answers one question: *given how long a person has been unloaded and which muscle is asked about, how much of that muscle is gone, and how much of the answer depends on anything else?* Two consequences follow from the wording. The prediction concerns a group, not a person, because the literature reports group means; and the framework is judged on whether it estimates the duration–muscle relationship honestly, not on how tightly a flexible model can be made to fit. [Figure 1](#) shows the chain from the search to the three tiers, and [Table 4](#) their roles.

4.2 Features

The covariate budget is set by the number of campaigns, not the number of rows. At roughly ten independent studies per study-level covariate [20], 32 campaigns permit about three study-level terms. Variables that vary *within* a campaign—muscle, arm, scan day—are far cheaper, because each campaign contributes its own internal contrast. The implemented feature set is:

- **Days of unloading at the scan**, entered through a nonlinear basis ([Section 4.3](#));

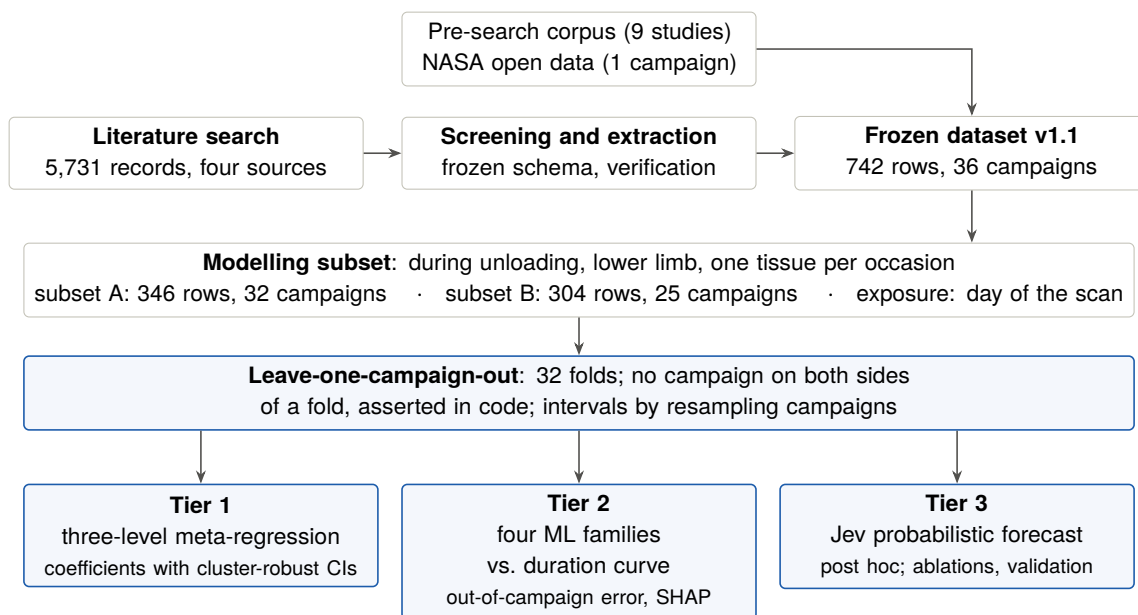


Figure 1. The framework. The dataset is frozen before any model is fitted; all three tiers read the same modelling subset, and tiers 2 and 3 are scored out of campaign against the same duration curve. Tier 1 uses cluster-robust inference on the campaign rather than cross-validation, because it estimates rather than predicts.

Table 4. The three tiers.

	Tier 1 — estimation	Tier 2 — comparison	Tier 3 — probabilistic forecast
Question	What is the duration–response, and which muscles deviate from it?	Does a flexible learner find structure the curve misses?	Can a model that reads each campaign’s description, and the other campaigns’ data, do better?
Method	Three-level meta-regression, nonlinear in the scan day	Ridge, random forest, SVR, gradient boosting	Language model returning a probability for each range of outcome
Validation	Cluster-robust inference on the campaign	Leave-one-campaign-out, nested tuning	Leave-one-campaign-out, ablations, recognition probe, validation battery
Reports	Coefficients, curve and ranking with 95% CIs	Out-of-campaign MAE, RMSE, pooled R^2	MAE, CRPS, log score, coverage of the most probable ranges
Status	Primary result; declared in advance	Declared in advance; allowed to fail	Added after the tier-2 null result; not pre-registered

- **Muscle family**, categorical: nine named families plus `whole_limb` in subset A;
- **Arm type**, binary (control or countermeasure). A dose ordinal or a countermeasure category would be fitting noise at this sample size: the 126 countermeasure rows are spread across nine named modalities, four of them with fewer than ten rows;
- **Modality and outcome**, collapsed to six levels (MRI volume, MRI CSA, CT CSA, DXA lean mass, ultrasound CSA, ultrasound thickness), because DXA lean mass and MRI volume do not measure the same thing;
- **Composite**, binary, so that the mixed granularity left by the resolution rule is visible to the model.

Age, sex, population, countermeasure modality, measurement site, tilt angle, nutrition control, body mass and the quality markers are recorded but not modelled: each is sparse, nearly constant, or almost perfectly

confounded with the campaign (for example, 347 of the 425 rows come from men-only groups). They are reserved for stratified sensitivity analyses. Provenance columns and `cohort_id` are never modelled; the campaign is the grouping variable, and the code asserts that it never appears as an input. The design document also lists the percentage-change estimator (Section 3.6) as a covariate; it is not in the implemented design matrix, and its effect is to be examined by sensitivity analysis S7 instead. [Draft note: decide whether to add the estimator flag to the model or to amend the design document.]

4.3 Tier 1: the three-level meta-regression

Model. For row i measuring muscle m in study s of campaign c ,

$$y_i = f(t_i) + \mathbf{x}_i^\top \boldsymbol{\beta} + u_c + v_{s(c)} + w_{m(c)} + \varepsilon_i, \quad \varepsilon_i \sim \mathcal{N}(0, \sigma_\varepsilon^2 / \omega_i), \quad (1)$$

where y_i is the percentage change, t_i the day of the scan, f one of the duration forms below, $\mathbf{x}_i$ the categorical and binary features, and $u_c \sim \mathcal{N}(0, \sigma_c^2)$, $v_{s(c)} \sim \mathcal{N}(0, \sigma_s^2)$ and $w_{m(c)} \sim \mathcal{N}(0, \sigma_m^2)$ random intercepts for the campaign, the study within the campaign and the muscle within the campaign. The last term is what makes the mixed granularity legitimate: repeated measurements of the same muscle in the same campaign are correlated, and the model is told so. The weights ω_i are the number of participants analysed, normalised to mean one, with missing values set to the median. This is the standard structure for pooled aggregate data with several effect sizes per study [16, 17], and it does not need the within-study covariances between muscles.

Estimation. Because every random effect is nested inside a campaign, the marginal covariance is block diagonal, with one block per campaign:

$$\mathbf{V}_c = \sigma_\varepsilon^2 \mathbf{W}_c^{-1} + \sigma_c^2 \mathbf{J}_c + \sigma_s^2 \mathbf{S}_c + \sigma_m^2 \mathbf{M}_c, \quad (2)$$

where $\mathbf{W}_c$ is the diagonal matrix of weights, $\mathbf{J}_c$ a matrix of ones, and $\mathbf{S}_c$ and $\mathbf{M}_c$ indicate pairs of rows from the same study and of the same muscle. The fixed effects are profiled out by generalised least squares at each evaluation, and the four variances are found by maximising the marginal likelihood on the log scale (L-BFGS-B). Maximum likelihood rather than REML is used because the duration forms do not share a design matrix and restricted likelihoods cannot be compared across them; the price is a mild downward bias in the variance components at 32 campaigns. The model was implemented directly in NumPy and SciPy rather than through a mixed-model library, which carries two levels natively but not the third or the sandwich below.

Inference. Confidence intervals and tests use a cluster-robust sandwich estimator clustered on the campaign [38, 39], on top of the random-effects structure:

$$\widehat{\text{Cov}}(\hat{\boldsymbol{\beta}}) = \kappa \mathbf{A}^{-1} \left(\sum_c \mathbf{X}_c^\top \mathbf{V}_c^{-1} \mathbf{r}_c \mathbf{r}_c^\top \mathbf{V}_c^{-1} \mathbf{X}_c \right) \mathbf{A}^{-1}, \quad \mathbf{A} = \sum_c \mathbf{X}_c^\top \mathbf{V}_c^{-1} \mathbf{X}_c, \quad (3)$$

with the small-sample factor $\kappa = \frac{G}{G-1} \cdot \frac{N-1}{N-p}$ for G campaigns, N rows and p fixed effects. Intervals use a t distribution with $G - 1$ degrees of freedom—31 in subset A—never the number of rows. The random effects model the correlation; the sandwich protects the inference if that model is misspecified, which at 32 clusters it may be.

Duration forms and the headline rule. Three forms of f are fitted and reported side by side:

- **logarithmic**, $f(t) = b \ln t$, the form used in a published synthesis [9], so that the two curves can be compared;

- **saturating exponential**, $f(t) = A(1 - e^{-t/\tau})$, whose two parameters are what a countermeasure planner wants: how fast the muscle gives up (τ) and how much it will eventually lose (A , read at the reference scenario). τ is profiled over the declared grid $\{7, 10, 14, 21, 28, 35, 45, 60, 90\}$ days by maximum likelihood and counted as one extra parameter;
- **restricted cubic spline** with three knots at the 10th, 50th and 90th percentiles of the scan day (10, 28 and 89 days) [19, 40], as a shape check.

The rule, declared before fitting, was: the saturating form is the headline unless another parametric form beats it by more than 4 AIC points; the spline is never the headline, and serves to show whether the parametric forms miss a shape in the data.

Reference levels and contrasts. Every contrast is read against a reference level and inherits its uncertainty, so the reference muscle family was chosen rather than inherited: knee extensors, with 86 rows across 22 campaigns in subset B, against the alphabetical default of dorsiflexors with 8 rows from 4 campaigns. The antigravity comparison—plantar flexors against dorsiflexors—is computed as a separate contrast from the coefficient difference, so that the choice of reference cannot change it. The modality reference is the alphabetical first level, CT cross-sectional area. The curve is drawn for one scenario: a control arm, the reference muscle family and modality, not a composite. Other scenarios shift the curve by their coefficients; they do not reshape it.

Fits. The model is fitted twice with the same specification: on subset A for the duration–response and on subset B for the muscle ranking, which is reported as the fitted change at day 60 for a control arm.

4.4 The duration-only baseline

The baseline is the simplest honest answer to “how much muscle is gone after t days”: percentage change as a function of the scan day and nothing else, in linear, logarithmic and saturating form, fitted by weighted least squares (weights ω_i) inside the same leave-one-campaign-out loop as everything else, with τ chosen from the same grid within each training fold. The best-scoring form is the reference against which tier 2 is judged.

4.5 Tier 2: the comparative machine-learning families

Four families were fitted on subset A, as promised in the accepted abstract: ridge regression (penalised linear regression) [41], random forest [22], support vector regression with a radial-basis kernel [24] and gradient boosting [23], all through scikit-learn [42]. The input is the feature set of Section 4.2 with the duration entered as the saturating basis at a fixed τ of 21 days; the flexible families can reshape it.

Validation. The fold unit is the campaign: 32 folds, each training on 31 campaigns and predicting one it has never seen. A random or paper-level split would place the same participants on both sides of the fold boundary and measure memorisation. The leakage guard is an assertion, not a convention: for every fold the code asserts that the training and test campaign sets are disjoint and that the campaign identifier is absent from the design matrix, and a test deliberately feeds it a random split to confirm that it fails.

Nested tuning. Hyperparameters were chosen inside each training fold by a grid search with five inner folds grouped by campaign, scored by mean absolute error, so that the outer fold sees none of the choice. The grids were small on purpose: ridge penalty $\alpha \in \{0.1, 1, 10, 100\}$ on standardised inputs; random forest (500 trees) minimum leaf size $\in \{2, 4, 8\}$ and feature fraction $\in \{0.5, 1\}$; SVR $C \in \{1, 10, 100\}$ and $\gamma \in \{0.05, 0.1, 0.5\}$ on standardised inputs; gradient boosting (300 trees) depth $\in \{1, 2\}$ and learning rate $\in \{0.03, 0.1\}$. The tier-2 families were fitted unweighted.

Metrics. Mean absolute error (MAE) and root mean squared error are computed within each held-out campaign and averaged with one vote per campaign, so that the largest campaign cannot dominate. The 95% interval is a percentile bootstrap over the 32 campaign-level errors (2,000 replicates, seed 20260918). R^2 is computed once over all pooled out-of-fold predictions rather than averaged over folds: most campaigns ran a single duration, so a duration-only model predicts one value across a held-out fold and scores below that fold’s own mean by construction, whatever the quality of the curve. The first baseline run returned about -8.5 per fold and $+0.06$ pooled from identical predictions.

Importance. SHAP values [27] were computed exactly (tree explainer) for the best-scoring family, recomputed inside every fold, and summarised as the share of folds in which each feature reaches the top three. Stability is reported first and magnitude second; nothing about importance is read causally.

4.6 Tier 3: a probabilistic language-model forecast

Status of tier 3

Tier 3 was added on 21 September 2026, after the tier-2 null result was known. It is not pre-registered. What could still be done was to declare the method and every bar before the first answer was requested, and to commit that declaration to the repository first; the same was done for the ablations and for the validation battery, each before its own first answer.

The question tier 2 leaves open is whether something that reads the *description* of a campaign—who the participants were, what they did, which muscle, which scanner—can extract what the coded columns of tier 2 could not. Tier 3 uses TypeSafe’s Jev (jev-1.13.0, pinned) [43], a model built to return a calibrated probability for each option of a fixed set given a described state. It is never asked for a number. For each row it is asked one question—*which of these ranges will the measurement fall in?*—and every numeric quantity is computed in code from the probabilities it returns.

Two arms.

- **Without history** (346 rows, 32 campaigns): the model sees nothing of the held-out campaign, as the tier-2 families do, and forecasts the percentage change from baseline over 2 pp ranges from -30 to $+4$, open at both ends. Reference: the logarithmic duration curve fitted to the training campaigns.
- **With history** (84 rows, 5 campaigns): the model also sees the held-out campaign’s scans strictly before the target day and forecasts the change since the previous scan, over 1 pp ranges from -14 to $+4$. References: no further change (*last scan*) and the curve’s step from the last scan’s day to the target day (*last scan plus curve step*). One campaign, BBR1, carries 39 of the 84 rows.

What the model is shown. A short background paragraph; the participants, protocol, countermeasure and measurement of the target row in words; the target day and its distance from the planned end; the duration curve fitted to the training campaigns; and as many of the training campaigns’ observations as fit a budget of 26,000 tokens, most relevant first (same muscle, then same family, then same kind of group, then nearest day). The budget was set from a measured rate of 2.3 characters per token so that state and question fit the model’s 32,000-token limit.

What it is never shown, by assertion. Any row of the held-out campaign in the reference tables; any scan of the held-out campaign on or after the target day; the target’s own outcome; and anything that names a paper, an author, a campaign or a registry. The last rule also keeps the model from recognising a published study by name and recalling its number. The assertions run on every request.

Baselines as distributions. Each baseline’s forecast distribution is its point prediction plus the residuals it made on the training campaigns, binned into the same ranges. The model and the baselines are therefore scored on identical footing for every metric.

Scores. The point forecast is the probability-weighted mean of the ranges’ representative values (the midpoint, or half a step beyond an open end). The continuous ranked probability score (CRPS) is computed as the ranked probability score [29] multiplied by the range width, so that it reads in percentage points and reduces to the absolute error for a forecast with no spread [28]. The log score is minus the log of the probability given to the range holding the truth, floored at 10^{-6} . The *most probable range* at level α is the shortest run of adjacent ranges whose probability reaches α ; its coverage and width are reported at 50% and 80%. Every metric is averaged within a held-out campaign and then across campaigns. The comparison statistic is the *paired gain*: the reference’s campaign MAE minus the model’s, averaged over campaigns, with a campaign bootstrap interval.

Bars, declared before the first answer.

- MAE: at least 15% relative improvement over the arm’s reference; otherwise report a null result.
- CRPS: at least 15% relative improvement, reported alongside the MAE and never instead of it.
- Coverage of the 80% range between 70% and 90%; otherwise the ranges may not be called 80% ranges.
- Leakage assertions hold for every request, and make forecast rebuilds every number from the committed answer cache without calling the service. Both blocking.

What tier 3 cannot rule out. Jev was pretrained on text that may include some of these papers. Withholding every name is the strongest guard available, but a model that has read a paper could still recognise it from a description plus a set of numbers. The ablations below were designed for that concern.

4.6.1 Ablations and the recognition probe

Three ablations on the arm without history each change one thing about what the model is shown, and are scored exactly as the first run was, against the same duration curve: *generic*, in which the target is described only by muscle, role, imaging method, kind of group and day—no participants, protocol text, planned length or measurement site, roughly the information tier 2 had; *scrambled reference*, in which the other campaigns’ values are shuffled among their rows; and *no reference*, with no curve and no other campaigns’ observations.

The *recognition probe* shows the model the target’s full description, exactly as the forecast sends it, and asks which of the 11 campaigns in subset A that have a proper name it comes from (AGBRESA, BBR1, BBR2-2, BRACE, LunHab, MEDES LTBR, NASA SPRINT, UTMB Campaign 3, PlanHab, VBR, WISE-2005), or none of them. Chance is one in twelve.

The reading rules were declared before any answer: a campaign is *recognised* if the model’s top choice is its true name on at least half of its rows; recognition is a *live explanation* of the gain if two or more of the three campaigns carrying most of the gain (NASA SPRINT, BBR2-2, MEDES LTBR) are recognised, or if the Spearman correlation between a campaign’s mean probability on its true name and its gain is 0.5 or more; the generic variant *keeps the gain* if its paired gain has an interval above zero and is at least half the full run’s (0.21 pp); and the scrambled variant *shows that the model reads the reference data* if its paired-gain interval reaches or falls below zero. If the generic variant keeps the gain and recognition is not live, the point forecasts may be quoted as a result.

4.6.2 The validation battery

A second set of runs, also declared before its first answer and also leave-one-campaign-out, asks whether the result depends on how the question was put and whether the history arm reads what it is given. On both arms, *reordered* shows the same reference rows in a different order and *shifted ranges* moves every range edge by half a step; neither changes what the model knows. On the history arm, the three ablations are repeated and a fourth, *scrambled history*, shuffles the held-out campaign’s earlier values among its earlier scans. A *repeat check* sends the first 20 requests of the arm without history again, past the cache. A result is *robust to presentation* if under both presentation checks its MAE moves by less than 10% and its paired gain keeps its sign (with an interval above zero on the arm without history); the model is *consistent* if all 20 repeats return identical probabilities; and the model *reads the earlier scans* if scrambling them worsens the error with an interval below zero. Tier 3 is called working rather than lucky if the arm without history is robust and consistent and the history arm reads what it is given.

4.7 Success criteria declared before modelling

The thresholds in [Table 5](#) were agreed in the project plan before any model was fitted. The first three were deliberately allowed to fail; the last three were blocking. They are assessed in [Section 6.10](#).

Table 5. Success criteria for tiers 1 and 2, declared before modelling (project plan, section 8).

Criterion	Target	If missed
Out-of-campaign MAE	≤ 4 pp	Report honestly; the dataset and framework remain the contribution
Best model vs. duration-only baseline	$\geq 15\%$ relative MAE improvement	Report the null result
SHAP top-3 stability	Same top three in $\geq 80\%$ of folds	Present importance as indicative only
Leakage audit	No campaign split across folds; no preprocessing outside a fold	Blocking
Reproducibility	One command regenerates every number and figure	Blocking
Physiological sign-off	Scientific lead signs off the fitted relationships	Blocking

4.8 Sensitivity analyses

Eleven sensitivity analyses were declared, each the same pipeline with one change ([Table 6](#)). S1, S2 and S4 were pre-registered as analyses that must be shown whatever they say. At the time of writing only S6 has been run ([Section 6.3](#)).

4.9 Software and reproducibility

The framework is a set of single-purpose Python modules (framework/) driven by one configuration file, framework/config.yaml, which holds every subset predicate, feature list, duration form, model grid, bootstrap setting and the random seed; no module contains a number that is not in the configuration. The core—loading, features, folds, baseline, evaluation and tier 1—needs only NumPy, pandas, SciPy and PyYAML, so that the primary result does not depend on the optional stack that produced the null result.

Table 6. Declared sensitivity analyses and their status on 21 September 2026. ★ = must be shown in the report.

	Change	What it tests	Status
S1★	Composite-first instead of component-first resolution	Whether the ranking is an artefact of granularity	Not run
S2★	Drop the MEDES 90-day campaign	Whether one campaign carries the duration–response	Not run
S3	Drop the two Berlin campaigns	The same for the other dominant group	Not run
S4★	MRI volume only	Whether mixing modalities changed the answer	Not run
S5	Exclude figure-derived and low-confidence rows	Whether digitisation is load-bearing	Not run
S6	Inverse-variance weights	Whether the weighting scheme matters	Run
S7	Each percentage estimator alone	Whether the two estimators agree	Not run
S8	Add mean age as a study-level covariate	Whether the older cohorts shift anything	Not run
S9	Sex-stratified, if the folds survive it	A question that will be asked	Not run
S10	Recovery rows with a phase term	Whether reconditioning can be modelled	Not run
S11	Rows with a stated measurement site, then by position	Whether slice position moves the estimate	Not run

The results reported here were produced with Python 3.13.0, NumPy 2.4.6, pandas 2.2.3, SciPy 1.17.0 and scikit-learn 1.8.0, and each results file records the dataset version, its hash and the commit it came from. `make all` regenerates every result from the frozen dataset. Tier 3 is rebuilt from the model’s answers committed in `results/forecast_cache/` (430 answers for the first run, 1,280 for the ablations and 1,195 for the validation battery), never by calling the service again. The test suite holds 223 checks in 19 files, including the deliberately failing leakage test; all 223 passed on 21 September 2026. The figures and tables of this report are regenerated from the results files by `report/make_figures.py`.

5 Results I: the corpus and the dataset

5.1 Search and screening

The four searches returned 5,731 records, of which 2,141 were duplicates (Figure 2). Of the 3,590 unique records, 2,493 were excluded on title and abstract, most often because the study did not use an unloading model (1,048) or reported no muscle outcome (830); 481 of the latter were real unloading campaigns that reported only gait, motor-unit or cardiovascular outcomes, and were kept as leads for the campaign map. Seventy-four records were included at title and abstract; 70 full texts were retrieved (30 as machine-readable XML, 40 as PDF) and four were not, two because they were paywalled and two because they exist only as conference abstracts. One of the latter still contributes: its abstract prints a complete result, which was extracted with low confidence [44]. The search contributed 42 studies, 572 rows and 29 campaigns to the dataset; 19 included studies publish their muscle results only as charts without a baseline value and could not be extracted.

A further 1,023 records—the remainder of the *maybe* set and 80 priority records that need a full text before a decision can be made—were not screened in the time available. They are not exclusions and are not reported as such. [Draft note: the full-text stage of the PRISMA table (`docs/literature-review/prisma_counts.md`) still reads “pending” for the number of studies included, and its count of 11 full-text exclusions carries reasons for only 5. The diagram shows the counts as recorded; they should be reconciled with the final extraction

before the report is circulated.]

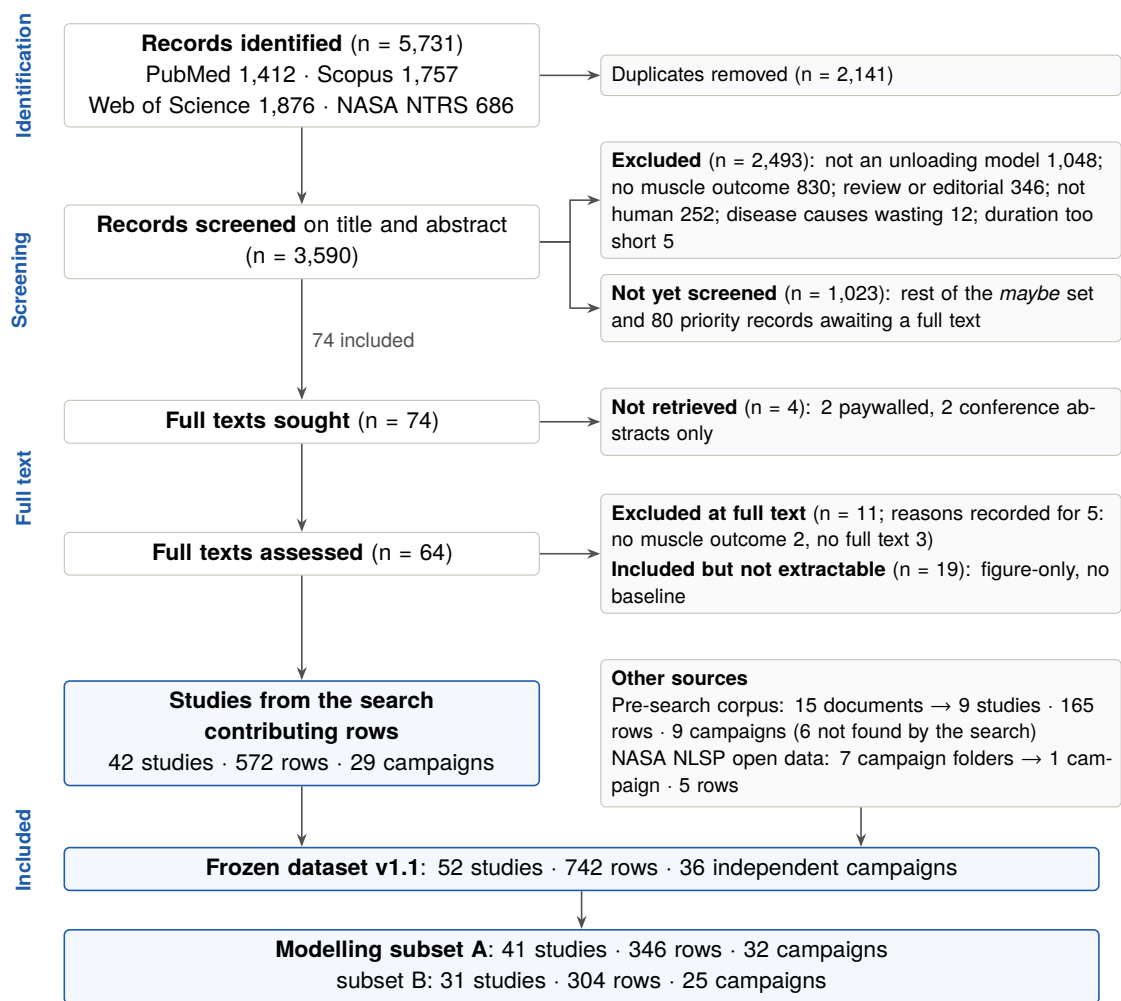


Figure 2. Flow of records from the search to the modelling subset, in the style of PRISMA 2020 [45]. The search was limited to publications from 2013 onwards; older work enters only through the pre-search corpus. Six of the nine corpus campaigns were not found by the search, because their papers predate the date limit.

5.2 The dataset

The frozen dataset (Table 7) holds 742 comparisons against baseline from 52 studies published between 1992 and 2026 and 36 independent campaigns. Most rows are MRI volumes of named muscles in young men during head-down tilt bed rest, but every other design, modality and population in the corpus is represented and marked. The dataset is strongly clustered: 15 campaigns contribute five rows or fewer, while the MEDES 90-day campaign contributes 299 rows (40%) to the full file. Figure 3 shows the 32 campaigns of the modelling subset, their planned lengths and the days on which they scanned; Section C lists all 36.

Origin of the rows. 572 rows (42 studies, 29 campaigns) come from the systematic search, 165 (9 studies, 9 campaigns) from the pre-search corpus and 5 (one campaign) from the NASA archive. Every literature row names its DOI, source file and page, table or figure; the NASA rows name their folder and the two columns they were computed from.

Quality markers. 674 rows carry high extraction confidence, 49 medium and 19 low; the 19 low-confidence rows are the ones read off figures. 25 partial rows lack a group size or a dispersion. 547 rows

Table 7. Characteristics of dataset_v1.1. Counts are rows unless stated.

Characteristic	Value
Rows (study $\times$ arm $\times$ muscle $\times$ occasion)	742
Studies	52, published 1992–2026
Independent campaigns (cohorts)	36
Distinct muscles or muscle groups	51; 276 rows are composites
Design	Head-down tilt bed rest 568; Horizontal bed rest 154; Spaceflight 14; Limb suspension 5; Dry immersion 1
Planned unloading	5–119 days on the ground; 180 days in spaceflight
Phase	during unloading 478; recovery 264
Arm	control 470; countermeasure 272
Modality	MRI 655; DXA 37; CT 30; ultrasound 20
Outcome	volume 594; CSA 100; lean mass 37; thickness 11
Source of the value	table 637; text 81; figure (digitised) 19; open data repository 5
Extraction confidence	high 674; medium 49; low 19
Population	healthy young 692; healthy older 42; healthy middle aged 8
Sex	men only 567; mixed 113; women only 40; NA 22

do not state which leg was imaged. No row has been double-extracted by a second person ([Section 3.7](#)).

5.3 The modelling subsets

[Table 8](#) walks the dataset down to the two modelling subsets. Restricting to rows measured during unloading removes 264 recovery rows, and the spaceflight and recovery-only campaigns with them; restricting to the lower limb removes 53 trunk-muscle rows; and the one-tissue-one-row rule removes 79 composite rows whose components were reported on the same occasion. Of the 346 rows in subset A, 246 are from control arms (30 campaigns) and 100 from countermeasure arms; 42 are whole-segment measurements; 251 are MRI volumes. Subset B removes the whole-segment rows and with them seven campaigns that report nothing else.

Table 8. From the frozen dataset to the modelling subsets. The predicates are declared in framework/config.yaml and applied in this order.

Step	Rows	Studies	Campaigns
All rows of dataset_v1.1.csv	742	52	36
Predicate 1: during unloading (phase = bed_rest)	478	46	32
Predicate 2: lower-limb muscles only	425	42	32
Predicate 3: one tissue per occasion — subset A	346	41	32
Named muscles only — subset B	304	31	25

5.4 A first look at the data

Before any model, the unadjusted control-arm means already show the pattern the models go on to quantify ([Table 9](#)): antigravity extensors lose about twice as much as flexors. By family, the plantar flexors lose most

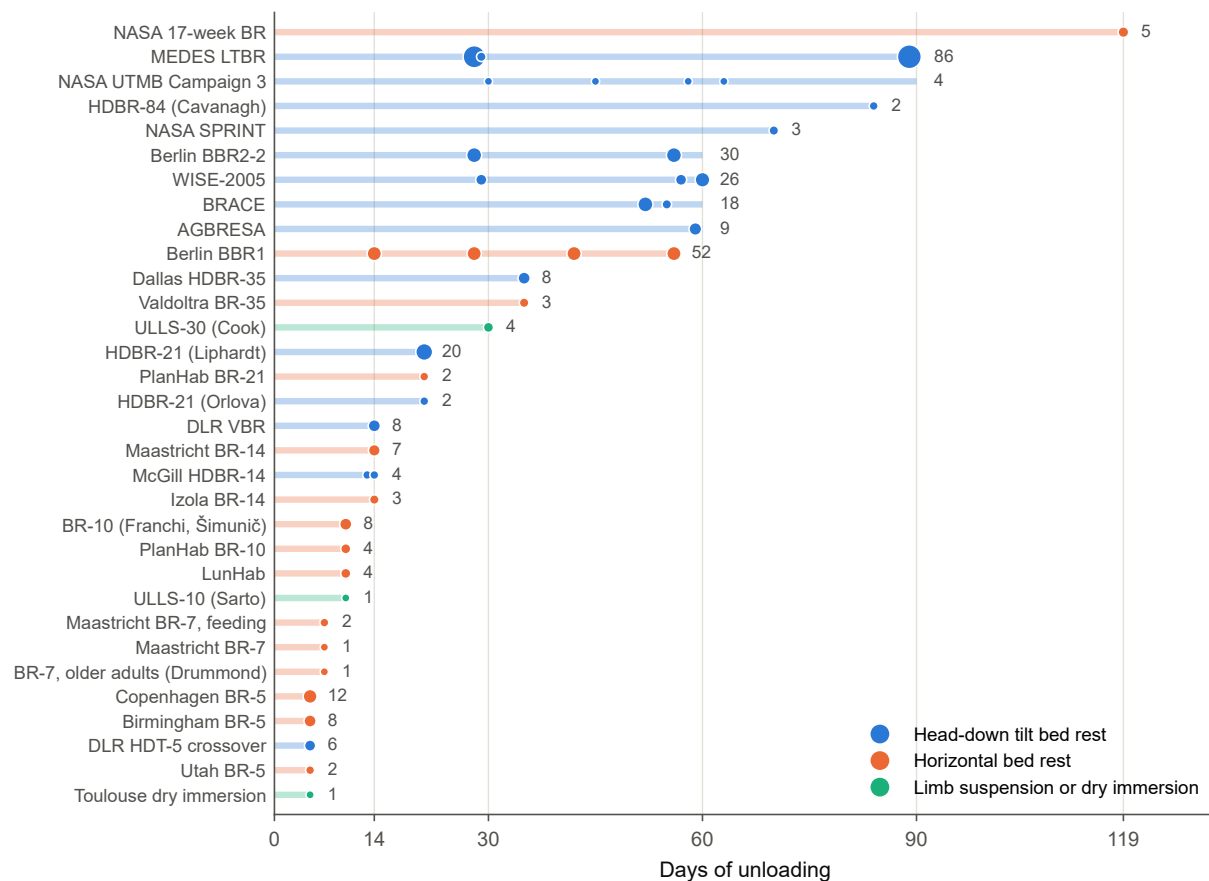


Figure 3. The 32 campaigns of the modelling subset, ordered by planned length. Each bar runs to the campaign’s planned length of unloading; each dot is a scan day, with area proportional to the number of rows measured that day; the number is the campaign’s rows in subset A. Most campaigns scanned only at the end; the long campaigns that scanned repeatedly carry most of the within-campaign information about time.

(−14.7%), followed by the evertors (−12.2%), knee extensors (−8.6%), knee flexors (−7.8%), dorsiflexors (−7.5%), hip adductors (−4.1%) and hip rotators (−1.3%). These means ignore duration, arm, modality and campaign, and are shown only as a sanity check on the muscle classification.

Table 9. Unadjusted mean percentage change of control-arm rows by functional class (lower-limb rows during unloading, before the one-tissue-one-row rule).

Functional class	Rows	Mean change (%)
Antigravity extensors	129	−12.0
Mixed function	137	−6.7
Flexors	33	−6.1

6 Results II: the three tiers

All results in this section are fitted on dataset_v1.1 with unloading measured at the day of the scan.

6.1 Tier 1: the duration–response

The shape of the curve. The three duration forms fit the 346 rows of subset A almost equally well (Table 10): the logarithmic, saturating and spline forms lie within 1.3 AIC points of one another, far inside the 4-point margin the headline rule required for another form to displace the saturating one. The data therefore establish that loss accumulates along a curved path, but not *which* curve. The spline shows no shape the parametric forms miss ($\Delta\text{AIC} = -0.2$). By the declared rule the saturating form is the headline.

Table 10. Tier 1, subset A (346 rows, 32 campaigns): the three duration forms. ΔAIC is relative to the headline saturating form; k counts fixed effects, the four variance components and, for the saturating form, τ . Intervals are 95% cluster-robust (t_{31}).

Form	AIC	ΔAIC	k	Duration term
Logarithmic, $a + b \ln t$	2087.0	+1.2	22	$b = -4.31$ pp per e -fold (-4.96 to -3.65); -2.99 pp per doubling (-3.44 to -2.53)
Saturating, $A(1 - e^{-t/\tau})$	2085.9	0.0	23	$\tau = 60$ days; eventual loss $A = -17.3\%$ (-20.9 to -13.7)
Restricted cubic spline	2085.7	-0.2	23	Knots at 10, 28, 89 days (shape check only)

How fast and how far. Under the logarithmic form, muscle size falls by 2.99 pp for every doubling of days (95% CI 2.53 to 3.44). Under the saturating form, the time constant is 60 days—the profile likelihood turns over inside the declared grid—and the eventual loss is -17.3% (-20.9 to -13.7) for the reference scenario. By day 119, the longest observation, the curve has covered 86% of the way to that plateau, so the plateau is an estimate at the edge of the data rather than far beyond it. Figure 4 shows the curve with the control-arm observations, and Table 11 its values at selected days.

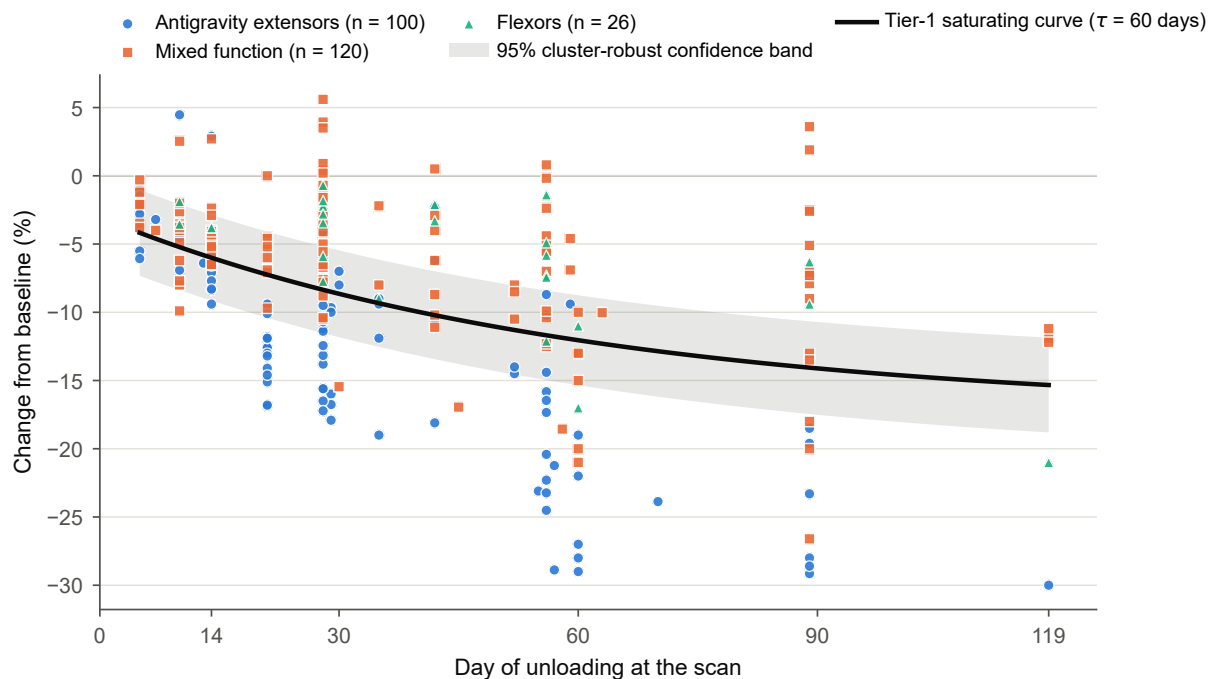


Figure 4. Duration–response. Points are the 246 control-arm rows of subset A (30 campaigns), placed at the day of the scan and coloured by functional class. The line is the tier-1 saturating curve for the reference scenario (a control arm, knee extensors, CT cross-sectional area, not a composite) with its 95% cluster-robust band. The scatter around the curve is dominated by which muscle was measured: antigravity extensors lie mostly below it, flexors and mixed muscles above.

Table 11. The tier-1 saturating curve at selected days for the reference scenario, with 95% cluster-robust confidence intervals.

Day of unloading	5	14	30	60	90	119
Fitted change (%)	−4.2	−6.0	−8.6	−12.0	−14.1	−15.3
95% CI	−7.3 to −1.0	−9.1 to −2.9	−11.8 to −5.5	−15.3 to −8.8	−17.5 to −10.7	−18.8 to −11.9

Where the variance sits. Table 12 decomposes the variance. In subset A, 54% is residual, 30% lies *between muscles within a campaign*, 11% between campaigns and 4% between papers reporting the same campaign. The 30% is the quantitative case for the second claim of the project: once duration is accounted for, which muscle is measured matters more than which campaign measured it.

Table 12. Tier 1 variance components (squared percentage points) and their share of the total.

Model	Campaign	Study within campaign	Muscle within campaign	Residual
Subset A, saturating form	2.84 (11%)	0.99 (4%)	7.72 (30%)	13.79 (54%)
Subset B, ranking model	4.64 (17%)	0.02 (0%)	8.15 (29%)	15.27 (54%)

Other terms. In subset A, countermeasure arms lose 3.43 pp less than control arms (95% CI 1.68 to 5.18, $p < 0.001$). Relative to CT cross-sectional area, DXA lean mass shows 4.75 pp less loss (0.05 to 9.45, $p = 0.048$) and ultrasound thickness 5.95 pp less (0.61 to 11.29, $p = 0.030$), while MRI volume differs by only 0.49 pp ($p = 0.72$); a curve drawn for MRI volume, the dominant measurement, would therefore sit about half a percentage point above the reference curve. The composite flag and the whole-segment family carry no detectable effect. All coefficients are listed in Section E.

6.2 Tier 1: muscle-specific vulnerability

The ranking model, fitted on the 304 named-muscle rows of subset B (25 campaigns), orders the muscle families by their fitted change at day 60 for a control arm (Table 13 and Figure 5). The plantar flexors lose most, −15.8% (−18.9 to −12.7), significantly more than the knee extensors (−5.69 pp, $p = 0.001$). The knee extensors, knee flexors and dorsiflexors form a middle group near −10%; the hip adductors, extensors and rotators lose significantly less than the knee extensors, the hip rotators least of all (−1.9%).

Table 13. Tier 1 muscle-family ranking, subset B (304 rows, 25 campaigns): fitted change at day 60 for a control arm, and contrast with the reference family (knee extensors). Intervals are 95% cluster-robust (t_{24}). Hip abductors and hip rotators each rest on a single campaign.

Family	Rows	Campaigns	Change at day 60, % (95% CI)	vs. knee extensors, pp (95% CI)	p
Plantar flexors	78	13	−15.8 (−18.9 to −12.7)	−5.69 (−8.95 to −2.42)	0.001
Dorsiflexors	8	4	−10.2 (−13.2 to −7.1)	−0.08 (−2.61 to 2.44)	0.945
Knee extensors	86	22	−10.1 (−13.6 to −6.6)	reference	—
Knee flexors	54	7	−9.8 (−12.1 to −7.6)	+0.28 (−2.36 to 2.92)	0.829
Hip abductors	6	1	−9.1 (−11.2 to −6.9)	+1.01 (−1.65 to 3.68)	0.440
Hip flexors	13	4	−6.8 (−9.9 to −3.6)	+3.33 (−0.13 to 6.80)	0.059
Hip extensors	8	3	−6.1 (−10.1 to −2.0)	+4.03 (0.79 to 7.27)	0.017
Hip adductors	39	4	−5.4 (−7.9 to −2.9)	+4.70 (2.12 to 7.28)	<0.001
Hip rotators	12	1	−1.9 (−3.9 to 0.2)	+8.22 (5.89 to 10.56)	<0.001

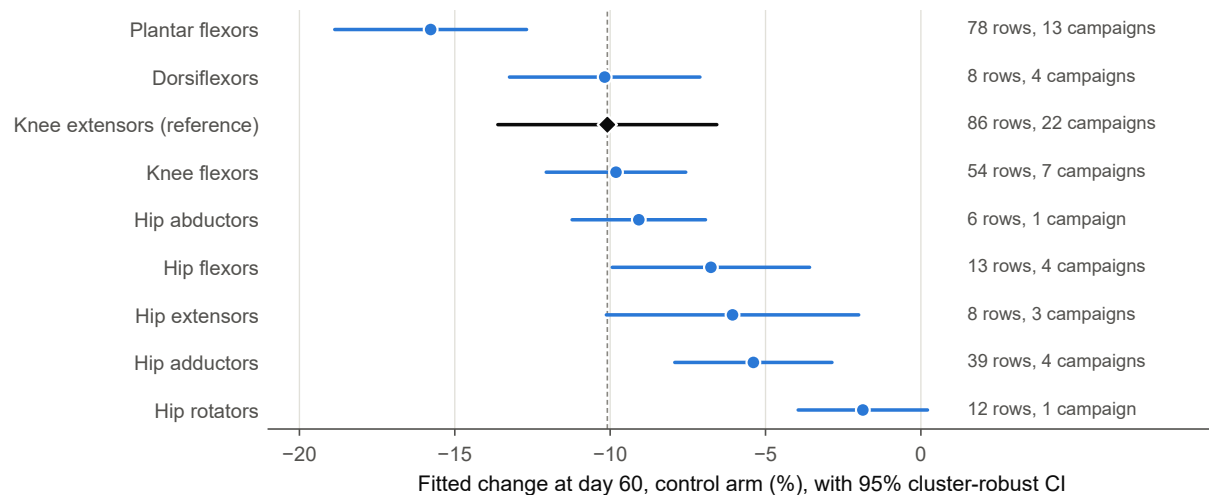


Figure 5. Muscle-family ranking at day 60 for a control arm, with 95% cluster-robust intervals, from the tier-1 ranking model on subset B. The dashed line marks the reference family. Rows and campaigns are shown for each family because two families rest on a single campaign.

The antigravity contrast. Because every contrast in Table 13 inherits the uncertainty of the reference, the comparison that carries the physiological claim is reported on its own: plantar flexors against dorsiflexors, the antagonist pair at the ankle, is -5.60 pp (95% CI -9.13 to -2.07 , $p = 0.003$), computed from the coefficient difference so that no choice of reference can change it. The ordering survives adjustment for duration, arm and modality.

Countermeasures. In the ranking model, countermeasure arms lose 3.73 pp less than control arms (95% CI 1.54 to 5.93 , $p = 0.002$). This is the first countermeasure effect the project has estimated rather than described; it pools every countermeasure type and says nothing about which works best.

6.3 Sensitivity analysis S6: does the weighting matter?

The primary analysis weights rows by the number of participants analysed. The classical alternative, inverse-variance weighting, needs a dispersion of the change, which only 161 of the 346 rows in subset A, from 8 of the 32 campaigns, carry; most papers print the spread of the baseline or nothing. Table 14 compares the two schemes on the same rows. The restriction to rows with a change dispersion moves the duration coefficient by 0.3 pp and the weighting scheme by another 0.9 pp, with overlapping intervals throughout: on this literature the scheme barely matters. The finding that belongs in the methods is the other one: inverse-variance weighting is unavailable as a primary scheme, because only eight campaigns carry what it needs.

Table 14. Sensitivity analysis S6. The duration coefficient is the saturating term of the tier-1 model; the error is out-of-campaign and campaign-weighted. From results/sensitivity.md.

Analysis	Weights	Rows	Campaigns	Duration coefficient (95% CI)	MAE (pp)
Primary: every row	participants	346	32	-14.25 (-16.40 to -12.11)	3.34
Rows with a dispersion of the change	participants	161	8	-14.51 (-15.61 to -13.42)	2.67
S6: the same rows	inverse variance	161	8	-13.60 (-16.69 to -10.50)	2.82

6.4 The duration-only baseline

Fitted inside the leave-one-campaign-out loop, the logarithmic curve is the best of the three baseline forms, with an out-of-campaign MAE of 3.16 pp (95% CI 2.44 to 3.93); the saturating form follows closely (3.19 pp) and the straight line is worst (3.40 pp), which is the evidence that the relationship is curved rather than assumed to be (Table 15). All three forms already meet the 4 pp target before any model family is fitted. Their pooled R^2 is low (0.07 to 0.14): duration alone predicts the average loss at a given day, but explains little of the row-to-row variation, because that variation is mostly which muscle was measured. Averaged over all muscles in subset A, the fitted saturating baseline has a time constant of 35 days and an eventual loss of -13.0% ; the whole-segment and composite rows pull that average towards the less affected tissue, which is why the tier-1 model with muscle terms, not the baseline, answers the question for a particular muscle.

6.5 Tier 2: the comparative machine-learning families

Tier 2 is a null result

No family beats the duration-only curve (Table 15 and Figure 6). The best, a random forest, scores 3.18 pp against the curve's 3.16 pp; the others are between 3.7% and 5.9% worse. The plan set the bar at a 15% relative improvement and pre-committed to reporting the outcome either way. This is rung B of the project's fallback ladder.

Table 15. Out-of-campaign error of the duration-only baseline in its three forms and of the four tier-2 families, subset A, 32 leave-one-campaign-out folds. MAE and RMSE are campaign-weighted; the interval is a campaign bootstrap; “vs. baseline” is the relative MAE change against the logarithmic curve (negative = worse); R^2 is pooled over out-of-fold predictions. The last column names the campaign with the largest error and that error.

Model	MAE (pp)	95% CI	RMSE	vs. baseline	R^2	Worst campaign (MAE)
Duration curve, linear	3.40	2.66 to 4.19	4.11	-7.7%	0.07	WISE-2005 (9.7)
Duration curve, logarithmic	3.16	2.44 to 3.93	3.88	reference	0.14	WISE-2005 (9.3)
Duration curve, saturating	3.19	2.46 to 3.98	3.93	-1.1%	0.12	WISE-2005 (9.2)
Random forest	3.18	2.50 to 3.91	3.71	-0.7%	0.34	HDBR-21 (Liphardt) (7.8)
Support vector regression	3.27	2.67 to 3.92	3.87	-3.7%	0.36	HDBR-21 (Liphardt) (7.9)
Ridge regression	3.28	2.59 to 4.02	3.93	-3.8%	0.27	WISE-2005 (8.6)
Gradient boosting	3.34	2.67 to 4.07	3.99	-5.9%	0.30	HDBR-21 (Liphardt) (7.8)

The informative part is panel b of Figure 6. Pooled R^2 rises from 0.14 for the curve to 0.27–0.36 once muscle identity enters: the flexible families do find real structure, but they do not convert it into a lower absolute error, because the residual is dominated by between-campaign differences that no column in the dataset explains. The worst folds show where: a 21-day campaign with site-resolved thigh measurements (HDBR-21, Liphardt; 7.8–7.9 pp) for three families, and WISE-2005, 60 days of bed rest in women only, for ridge regression (8.6 pp).

Importance. SHAP values on the random forest identify two stable features (Table 16): the duration term reaches the top three in 29 of 32 folds (91%) and the plantar-flexor indicator in 28 (88%). The third place alternates between the countermeasure arm (59%) and the composite flag (53%), so the declared criterion—the same top three in 80% of folds—is missed. The flexible models lean on exactly the two variables the tier-1 model estimates with intervals, which is the consistency the importance analysis was meant to check; it is not evidence of cause.

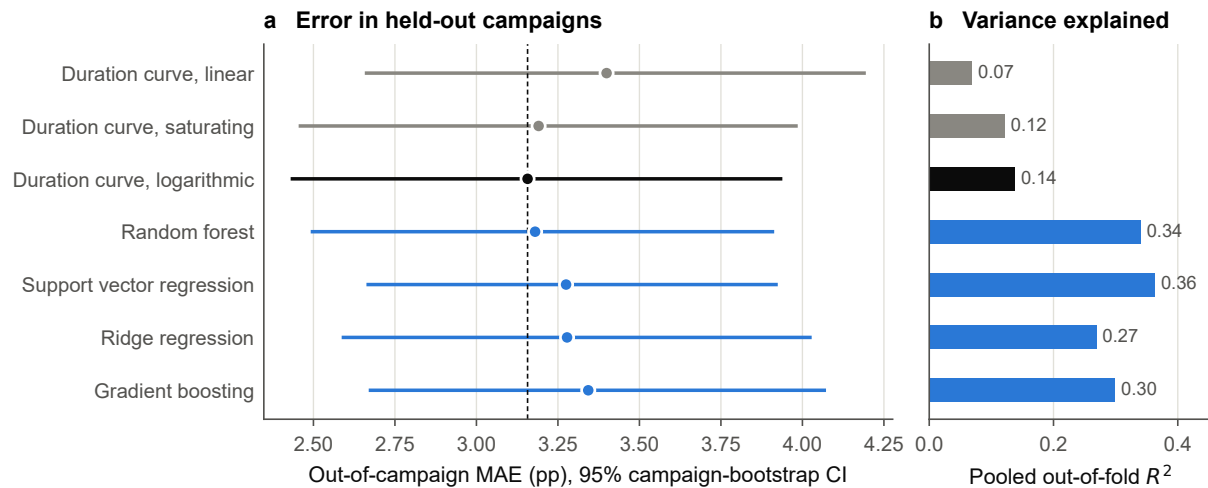


Figure 6. Tier 2 against the duration curve. (a) Out-of-campaign MAE with 95% campaign-bootstrap intervals; the dashed line is the logarithmic curve. (b) Pooled out-of-fold R^2 . The flexible families explain two to three times more variance than the curve, yet do not convert it into lower error.

Table 16. Stability of SHAP importance for the random forest across the 32 folds: how often each feature reaches the top three, and its mean absolute SHAP value. Features never in the top three are omitted.

Feature	Folds in top three	Share	Mean SHAP
$1 - e^{-t/\tau}$	29 of 32	91%	3.92
Plantar flexors	28 of 32	88%	1.71
Countermeasure arm	19 of 32	59%	1.04
Composite muscle	17 of 32	53%	0.87
Ultrasound thickness	2 of 32	6%	0.18
MRI CSA	1 of 32	3%	0.21

6.6 Tier 3: the probabilistic forecast

The first run sent 430 requests, the largest of 26,390 input tokens. Table 17 reports both arms against their references on identical metrics.

Table 17. Tier 3 against its references. MAE and CRPS in percentage points, campaign-weighted, with a campaign-bootstrap interval for the MAE; the log score is per row (lower is better); coverage and mean width (pp) of the most probable 50% and 80% ranges. [†] the arm's reference.

Model	MAE	95% CI	CRPS	Log score	50% range	80% range	R^2
<i>Without history: percent change from baseline (346 rows, 32 campaigns)</i>							
Jev (TypeSafe)	2.71	2.12 to 3.33	2.05	2.84	39% / 2.8	62% / 5.4	0.43
Duration curve [†]	3.13	2.37 to 3.92	2.49	2.29	62% / 8.0	90% / 15.4	0.14
<i>With history: change since the previous scan (84 rows, 5 campaigns)</i>							
Jev (TypeSafe)	2.85	2.19 to 3.50	2.21	4.88	22% / 1.3	38% / 2.8	0.19
Last scan	3.66	2.91 to 4.42	2.44	4.69	56% / 4.2	69% / 9.0	-0.22
Last scan plus curve step [†]	3.21	2.58 to 3.87	2.16	3.90	50% / 4.9	77% / 8.8	0.02

Without history. Jev’s error is 2.71 pp against the curve’s 3.13 pp.¹ Paired by campaign, the model’s error is lower by 0.42 pp (95% CI 0.13 to 0.73), and it is closer in 20 of the 32 campaigns (Figure 7). This is the first advantage over the duration curve in the project whose paired interval excludes zero, and the pooled R^2 of 0.43 is about three times the curve’s (Figure 8). Against the declared bars, however, the MAE improvement is 13.3%, short of the 15% required; the CRPS improvement, 17.7%, meets its bar; and the 80% ranges contain the truth only 62% of the time, so they may not be called 80% ranges. The log score says the same from the other side: it is worse than the curve’s (2.84 against 2.29) despite the better CRPS, because when the model is wrong it has placed almost no probability where the truth landed.

With history. Forecasting the change since the previous scan on 84 rows from five campaigns, Jev reaches 2.84 pp against 3.21 pp for the last scan plus the curve’s step, a paired gain of 0.37 pp (0.03 to 0.75) in four of the five campaigns. The MAE improvement (11.5%) misses its bar, the CRPS is 2.2% worse than the reference, and the ranges are badly overconfident: 2.8 pp wide, they contain the truth 38% of the time. With five campaigns, every interval on this arm rests on a bootstrap over five values and is indicative only.

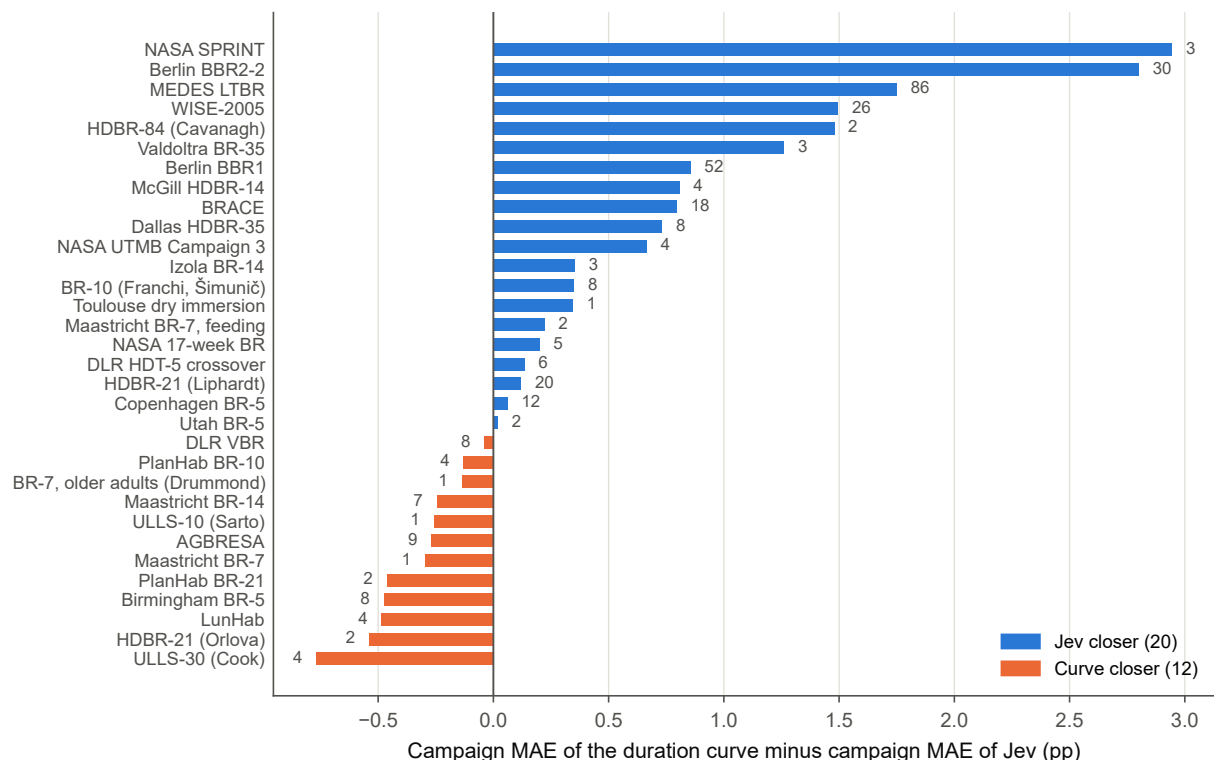


Figure 7. Tier 3 without history: the duration curve’s campaign MAE minus Jev’s, for each held-out campaign (positive = Jev closer). The number beside each bar is the campaign’s rows. More than half of the total gain comes from three long, multi-muscle MRI campaigns—NASA SPRINT, Berlin BBR2-2 and MEDES LTBR—which are also among the most published campaigns in the field.

6.7 Tier 3: is the gain signal or recognition?

Because more than half of the gain came from three much-published campaigns, the ablations and the recognition probe were declared and run (1,280 new answers). Table 18 and panel a of Figure 10 report them.

¹The curve scores 3.13 pp here and 3.16 pp in Table 15 because here its point is the mean of its forecast distribution, the same definition used for the model’s point.

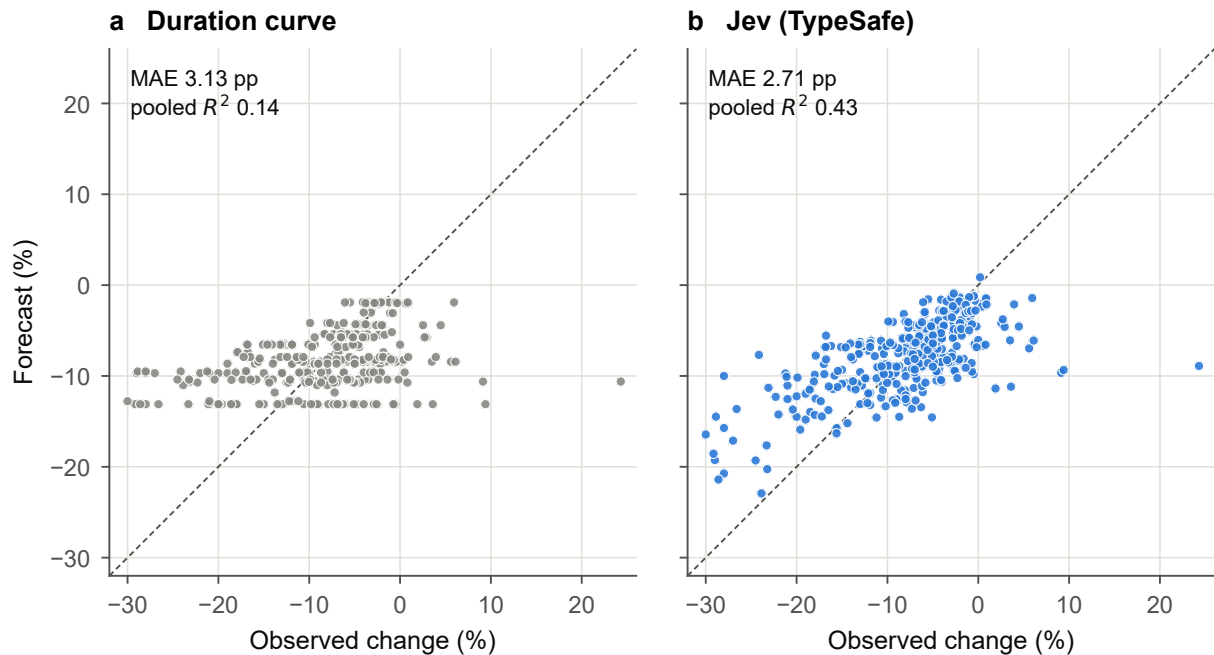


Figure 8. Forecast against observed change for all 346 rows of subset A, each forecast made with its campaign held out. (a) The duration curve predicts nearly one value per scan day. (b) Jev follows the between-muscle spread, which is where its gain comes from; both regress towards the mean at the extremes. Dashed line: identity.

Table 18. Tier 3 ablations on the arm without history (346 rows, 32 campaigns). The paired gain is over the duration curve; the change is relative to the full run (negative = worse). Intervals are campaign bootstraps.

Variant	MAE	CRPS	80% cover	Paired gain, pp (95% CI)	Change vs. full, pp (95% CI)	Closer
Full state (as run)	2.71	2.05	62%	+0.42 (0.13 to 0.73)	—	20 of 32
Generic description	2.83	2.25	54%	+0.30 (0.01 to 0.62)	-0.11 (-0.24 to 0.02)	20 of 32
Reference values shuffled	4.37	3.36	58%	-1.25 (-1.94 to -0.59)	-1.66 (-2.32 to -1.07)	7 of 32
No reference data	5.85	3.54	86%	-2.72 (-3.58 to -1.76)	-3.14 (-3.92 to -2.29)	6 of 32
Duration curve (reference)	3.13	2.49	90%	—	—	—

Against the reading rules.

- **The generic variant keeps the gain:** 0.30 pp (0.01 to 0.62), at least half of the full run's 0.42 pp with an interval above zero; its difference from the full run, -0.11 pp, has an interval that includes zero. The gain does not need anything that identifies a study.
- **The model reads the reference data:** with the other campaigns' values shuffled, the gain becomes a loss of 1.25 pp; without any reference data the model is far worse than the curve (5.85 against 3.13 pp).
- **Recognition is not a live explanation** (Figure 9). The model places a mean probability of 0.14 on the true campaign, against 0.08 by chance, and names it first on 19% of rows. Three campaigns pass the recognition bar—NASA SPRINT and AGBRESA (named first on 67% of their rows) and VBR (50%)—but of the three that carry most of the gain only SPRINT is recognised; for BBR2-2 and MEDES LTBR the model's commonest answer is “none of these”. Recognition and gain correlate at $\rho = 0.37$ ($p = 0.26$, 11 campaigns).

By the declared table this is the first case: *tier 3's point forecasts may be quoted as a result*, with the caveats of Section 6.6. The campaign-level errors point the same way. A model recalling published numbers should do best without reference data on the campaigns it can name; it does not, being worse than

the curve on AGBRESA (6.13 against 2.95 pp) and VBR (5.73 against 0.95 pp). SPRINT is the exception (4.80 against 5.92 pp) and the one place recognition may contribute, but the generic variant, which gives the model nothing to recognise SPRINT by, still beats the curve there (3.79 against 5.92 pp). The one campaign no paper reports, UTMB Campaign 3, computed from NASA’s raw archive, also favours the model (0.67 pp gain, four rows).

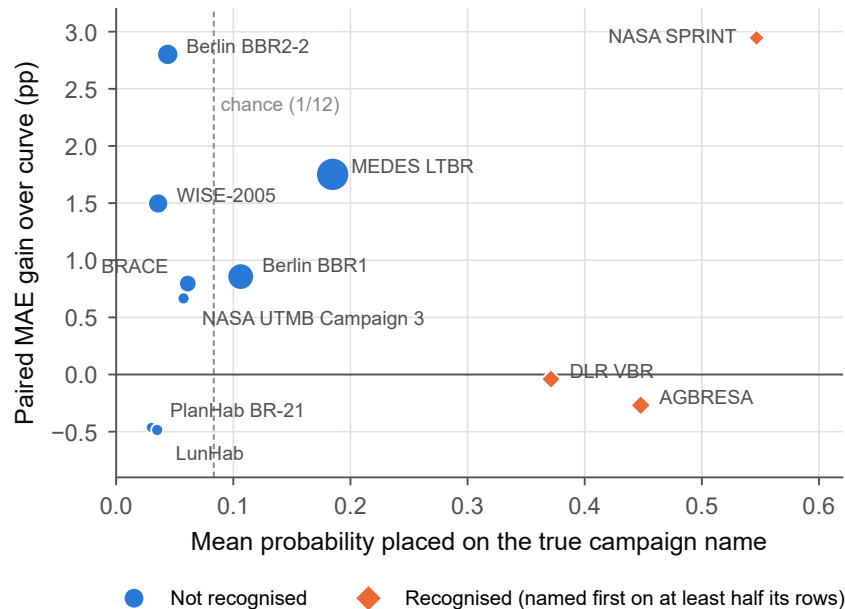


Figure 9. Recognition probe, 11 campaigns with proper names. Horizontal: mean probability the model places on the true campaign name when shown the description the forecast uses; vertical: the campaign’s paired gain over the curve. Marker area is proportional to rows. Two of the three campaigns carrying most of the gain (BBR2-2, MEDES LTBR) are not recognised.

6.8 Tier 3: does it hold up?

The validation battery added 1,195 answers and 20 repeats past the cache (Table 19, Figure 10).

Against the reading rules.

- **Robust to presentation: yes, on both arms.** Without history the MAE moves by 3% under reordering and 1% under shifted ranges, and the gain keeps an interval above zero under both (0.34 and 0.45 pp). On the history arm both moves are 2% and the gain keeps its sign.
- **Consistent: no, by the letter of the rule.** None of the 20 repeated requests returned identical probabilities. The probabilities moved by 0.034 on average (0.09 at most), the forecasts by 0.17 pp on average (0.48 at most), the most likely range held in 18 of 20, and the error on the repeated rows went from 2.456 to 2.479 pp. Run-to-run variation is of the order of the differences between presentation runs and far below the 0.42 pp gain.
- **Reads the earlier scans: no.** Shuffling the held-out campaign’s earlier values leaves the error where it was (+0.06 pp, interval across zero).
- **Reads the other campaigns on the history arm: not shown.** Both reference ablations are worse in their point estimates, by 0.13 and 0.52 pp, but on five campaigns neither interval clears zero.
- **The earlier scans help—strongly, but not through the model.** On the same 84 rows the model’s error falls from 5.83 to 2.84 pp when it forecasts the change since the last scan rather than the level (paired 2.98 pp, 1.54 to 5.51), and the baselines’ from 7.27 to 3.21 pp (4.06 pp, 2.52 to 5.99). What the

Table 19. Tier 3 validation battery, leave-one-campaign-out throughout. MAE in pp for Jev and for the arm’s reference; paired gain over the reference and change from the full run with 95% campaign-bootstrap intervals. On the history arm each interval rests on five campaigns.

Run	Jev MAE	Reference MAE	Paired gain, pp (95% CI)	Change vs. full, pp (95% CI)
<i>Without history (346 rows, 32 campaigns; reference: duration curve)</i>				
Full state (as run)	2.71	3.13	+0.42 (0.13 to 0.73)	—
Reference rows reordered	2.79	3.13	+0.34 (0.07 to 0.62)	−0.08 (−0.20 to 0.05)
Ranges shifted half a step	2.68	3.13	+0.45 (0.15 to 0.78)	+0.03 (−0.03 to 0.09)
<i>With history (84 rows, 5 campaigns; reference: last scan plus curve step)</i>				
Full state (as run)	2.85	3.21	+0.37 (0.03 to 0.76)	—
Generic description	2.87	3.21	+0.34 (0.10 to 0.59)	−0.03 (−0.23 to 0.14)
Reference values shuffled	2.98	3.21	+0.24 (−0.22 to 0.77)	−0.13 (−0.43 to 0.10)
No reference data	3.37	3.21	−0.15 (−1.48 to 0.94)	−0.53 (−1.57 to 0.48)
Earlier scans shuffled	2.78	3.21	+0.43 (0.08 to 0.78)	+0.06 (−0.14 to 0.28)
Reference rows reordered	2.78	3.21	+0.44 (0.07 to 0.90)	+0.07 (−0.06 to 0.17)
Ranges shifted half a step	2.90	3.21	+0.31 (0.04 to 0.58)	−0.05 (−0.27 to 0.10)

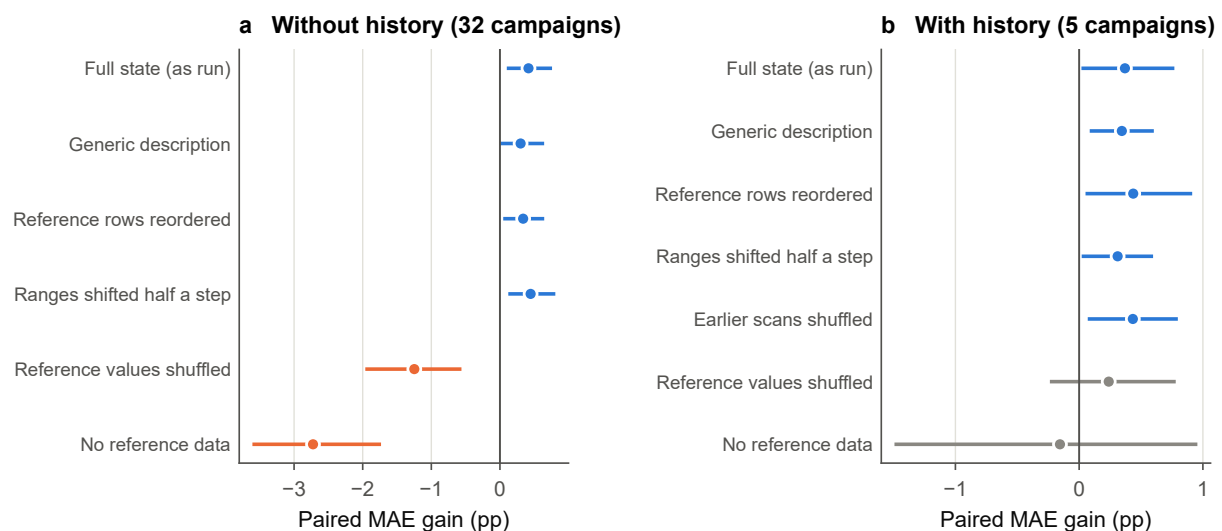


Figure 10. Paired MAE gain of Jev over each arm’s reference under every ablation and presentation check, with 95% campaign-bootstrap intervals. Blue: interval above zero; orange: below zero; grey: spans zero. (a) Without history the gain survives every change of presentation and the removal of identifying detail, and collapses when the reference data are shuffled or removed. (b) With history the gain is unchanged when the campaign’s own earlier scans are shuffled.

earlier scans supply is the anchor, and the code supplies it by adding the forecast change to the last scan.

Of the three conditions for calling tier 3 working rather than lucky, the one that carries the quoted number holds: the result without history survives both presentation checks and its run-to-run variation is far smaller than the gain. The other two fail and are reported as failing. The tier-3 number is therefore quoted from the arm without history, with the note that repeated runs differ slightly; nothing is claimed about the model using a campaign’s earlier scans.

6.9 The effect of correcting the time axis

Table 20 records what changed when tiers 1 and 2 moved from the campaign’s planned length to the day of the scan ([Section 3.12](#)). The fit improved by about 65 AIC points on identical rows, the plainest evidence that the scan day is the right exposure; the time constant moved from the edge of the declared grid to its interior; and the share of variance between muscles rose. What matters for the conclusions did not change: the duration forms remain indistinguishable, the ranking keeps its order, and no tier-2 family beats the curve. Tier 3 was fitted on the scan day from the start.

Table 20. Results fitted on the campaign’s planned length (first runs, superseded) and on the day of the scan (reported). Values for the planned length are as recorded in `framework/DESIGN.md`, section 9.4.

Result	Planned length	Day of the scan
Tier 1, saturating form, AIC on the same 346 rows	2152.3	2085.9
Residual variance	17.7	13.8
Loss per doubling of days, logarithmic form (pp)	−2.41 (−3.26 to −1.56)	−2.99 (−3.44 to −2.53)
Saturating time constant	90 days, at the edge of the grid	60 days , inside it
Eventual loss, saturating form (%)	−19.9 (−24.8 to −15.0)	−17.3 (−20.9 to −13.7)
Share of variance between muscles within a campaign	23%	30%
Baseline (logarithmic), out-of-campaign MAE	3.21 pp	3.16 pp
Baseline pooled R^2	0.03	0.14
Best tier-2 family against the baseline	SVR, −0.4%	Random forest, −0.7%
Tier-2 pooled R^2	0.10–0.22	0.27–0.36
Features stable in the top three (≥80% of folds)	duration only, at 69%	duration (91%), plantar flexors (88%)

6.10 Assessment against the declared criteria

Table 21 sets every criterion declared before modelling, and every bar declared for tier 3, against its outcome. Of the six criteria for tiers 1 and 2, the three that were allowed to fail split one met and two missed; of the three blocking criteria, two are met and the physiological sign-off is outstanding.

Table 21. Assessment against the criteria of [Table 5](#) and the tier-3 bars.

Criterion	Outcome	Evidence
Out-of-campaign MAE ≤ 4 pp	Met	Every baseline form and every family, 3.16–3.40 pp
Best model $\geq 15\%$ better than baseline	Not met	Best family -0.7% ; null result reported
SHAP top-3 stability $\geq 80\%$	Not met	Top two stable (91%, 88%); third place 59%
Leakage audit	Met	Disjointness asserted in every fold; the assertion is tested against a bad split
Reproducibility	Met	One command rebuilds every result; 223 tests pass; tier 3 rebuilt from its cache
Physiological sign-off	Outstanding	Muscle classification and fitted relationships await the scientific lead
Tier 3, MAE $\geq 15\%$ better	Not met	13.3% without history, 11.5% with
Tier 3, CRPS $\geq 15\%$ better	Met without history	17.7%; with history 2.2% worse
Tier 3, 80% coverage in 70–90%	Not met	62% and 38%: ranges overconfident
Tier 3, leakage and reproducibility	Met	Assertions on every request; cache rebuild without API calls

7 Discussion

7.1 Principal findings

Four statements are supported by the evidence, listed in order of how well it supports them.

1. **Muscle loss during bed rest follows a curved, not linear, path:** about 3 pp per doubling of days (95% CI 2.5 to 3.4), with the reference curve at -12% by day 60 and -15% by day 119. The corpus cannot say *which* curve: logarithmic, saturating and spline forms are indistinguishable.
2. **Which muscle is measured matters more than anything else in the dataset.** Thirty per cent of the variance lies between muscles within a campaign; plantar flexors lose 5.7 pp more than knee extensors and 5.6 pp more than their antagonists, the dorsiflexors; antigravity extensors lose roughly twice what flexors lose. The flexible models of tier 2 lean on the same two variables.
3. **At 32 campaigns, tabular machine learning adds nothing to a simple curve.** The four families find the muscle structure—pooled R^2 doubles or triples—but do not convert it into lower out-of-campaign error. That is a statement about how much signal the published literature contains, not about the algorithms.
4. **A model that reads the other campaigns’ data in context adds a little.** The language-model forecast is 0.42 pp more accurate than the curve, survives the removal of study-identifying detail and every presentation check, and is not explained by recognition of published campaigns. It was not pre-registered, it misses its declared bar and its ranges are overconfident; it is a finding to report, not the headline.

7.2 The duration–response

The clearest single result is negative about shape and positive about magnitude. A straight line in days is the worst of the three baseline forms, and the curvature is real: roughly 40% of the loss the reference

curve reaches by day 119 has already occurred by day 14 (-6.0% of -15.3%). What the data cannot do is choose between a curve that keeps falling with the logarithm of time and one that approaches a plateau. The 32 campaigns are concentrated at 5–21 and 56–90 days, with a single campaign at 119 days, and both forms pass through that support equally well. The saturating form was kept as the headline by a rule declared in advance, because its parameters are interpretable—a time constant of 60 days and an eventual loss of about 17%—not because the data prefer it. For planning purposes the honest summary is the curve and its band over the observed range, not the asymptote.

Two properties of the curve should be kept in view when quoting it. It is drawn for one scenario: a control arm, knee extensors, measured as CT cross-sectional area. For MRI volume it sits about half a percentage point higher, and for plantar flexors about 5.7 pp lower. And it is fitted to group means: it describes what a group of similar people would lose on average, not what an individual will lose.

7.3 Muscle-specific vulnerability: mechanical role

The ranking reproduces, as adjusted coefficients with intervals, the selective vulnerability reported by muscle-resolved bed-rest studies [3, 4, 9]: the plantar flexors are the most affected family and the hip rotators and adductors the least. The corpus also offers a partial test of the two usual explanations. Tibialis anterior, the principal dorsiflexor, is slow-fibre dominant like the soleus [10] but does not carry body weight; if fibre type governed atrophy, the dorsiflexors should lose like the plantar flexors. They do not: they lose like the knee extensors (-10.2% against -10.1% at day 60) and 5.6 pp less than the plantar flexors. The comparison rests on eight dorsiflexor rows from four campaigns and on a classification that has not yet been signed off, so it is suggestive rather than decisive, but it points towards mechanical role.

The countermeasure effect, 3.7 pp less loss in countermeasure arms, pools every countermeasure type, from flywheel exercise to nutrition. It shows that the dataset can detect a countermeasure effect at all; it cannot say which countermeasure works, which would need more campaigns per modality than the literature provides.

7.4 Why flexible learners did not help

The tier-2 null result is the most important methodological finding of the project, and the explanation is the sample size. There are 346 rows, but 32 independent campaigns, and the error that matters is the error on a campaign the model has never seen. Most of what distinguishes one campaign from another—its imaging protocol, its population, the exact definition of its muscles, its countermeasure dose—is either not recorded in a way that can be modelled or is perfectly confounded with the campaign itself. The flexible families learn the muscle structure, which raises pooled R^2 , but the part of the residual that decides out-of-campaign error is between-campaign, and no column explains it. With roughly a tenth of the sample size at which flexible learners begin to behave well [25], this is the expected outcome, and it was prepared for in advance.

The answer this gives to the question “why not simply use machine learning?” is a better one than any accuracy figure would have been: because at this sample size the unit of evidence is the campaign, and the literature holds 32 of them. It also explains why the meta-regression, and not a predictive model, is the right primary tool: it estimates exactly the structure the learners found, with intervals, and it does not need to predict unseen campaigns to be useful.

7.5 What tier 3 does and does not show

The language-model forecast is the only method in the project whose advantage over the curve has a paired interval that excludes zero. The ablations establish what the gain does not need—anything that identifies

a study—and that it depends on the reference values: shuffle them and the gain becomes a loss. The recognition probe finds that the model can name some campaigns, but not the ones carrying the gain. The likeliest reading is that the model does a kind of case-based reasoning that tier 2 could not: it picks out the reference rows that match the target—same muscle, same kind of group, a nearby day—and weighs them, where tier 2 received the same information as one-hot columns and a duration basis. That is an interpretation, not a tested mechanism.

The limits are as clear as the gain. The method was added after the null result; it misses the 15% bar declared for it; its ranges are overconfident, so its uncertainty cannot be quoted; its answers vary slightly between identical requests; and on the history arm it does not use the campaign’s earlier scans at all—the benefit of history comes entirely from the anchor the code supplies. Pretraining exposure to some of these papers cannot be excluded by any cross-validation.

7.6 Implications for mission planning

The results support three statements a countermeasure planner can use, at group level and within the observed range of 5 to 119 days: that most of the loss seen by four months has accumulated within the first weeks; that the calf’s antigravity muscles are the ones to protect first; and that countermeasure arms in the literature lose measurably less than controls. They do not support a prediction for an individual astronaut, a statement about actual microgravity, or a number for day 180. Extrapolation to a six-month transit was specified but has not been run; when it is, it will be computed from the tier-1 curve only, reported as a prediction interval drawn on the same axes as the data, and accompanied by the statement that the longest observation is 119 days and that the remainder is an assumption about curve shape.

7.7 Reconciliation with the accepted abstract

The accepted abstract was written before the literature search and is the contract for the presentation. The results confirm its central claims and correct several of its numbers (Table 22). None of the corrections weakens the claims; most strengthen the evidence behind them.

8 Limitations

The limitations are stated in full; several are properties of the literature rather than of the analysis.

1. **Small effective sample.** The modelling subset holds 32 independent campaigns. Every interval is wide accordingly, and the covariate budget is about three study-level terms.
2. **Clustering and dominance.** One campaign (MEDES LTBR) supplies 40% of the full dataset and a quarter of the modelling subset; 15 campaigns contribute five rows or fewer. Campaign weighting and campaign-level validation mitigate this, but the must-show sensitivity analysis that drops the largest campaign (S2) has not been run.
3. **Search coverage.** The search starts in 2013; older work enters only through a convenience corpus of nine papers. Embase was not searched, 1,023 records were not screened in the time available, and 19 included studies report their results only as charts without a baseline.
4. **No human double extraction.** The extraction was verified against the sources with AI assistance, and the pre-search corpus rows were not verified by a second person at all.
5. **Heterogeneous measurements in one target.** MRI volume, CT and MRI cross-sectional area, DXA lean mass and ultrasound thickness are treated as one target with a modality term; the MRI-only sensitivity analysis (S4) is outstanding. The measurement site along a muscle, which matters [4], is

Table 22. The accepted abstract against the results.

Abstract	Result
“15 bed-rest and HDBR studies (1992–2025)”	52 studies (1992–2026) from 36 independent campaigns; the modelling subset holds 41 studies from 32 campaigns
“unloading durations of 14–119 days”	Planned durations and scan days both span 5–119 days
Variables include participant characteristics	Recorded, but not modelled: age and sex are almost perfectly confounded with the campaign and are reserved for sensitivity analyses S8 and S9
Linear regression, random forest, SVR, gradient boosting	As stated; linear regression is penalised (ridge)
Leave-one-study-out cross-validation	Leave-one- <i>campaign</i> -out: stricter, because papers from one campaign share participants
SHAP-based interpretation	Done for the best family; the top two features are stable, the third is not
“consistent relationship between duration and loss”	Confirmed: curved, about –3 pp per doubling of days
Soleus, gastrocnemius and triceps surae most susceptible, “losses approaching 20–30% during prolonged bed rest”	Plantar flexors are the most affected family, –15.8% at day 60 by the model; individual group means reach –30% at the longest durations, but the modelled averages do not
“Exercise countermeasures consistently associated with reduced muscle loss”	Countermeasure arms lose 3.7 pp less (95% CI 1.5 to 5.9), all countermeasure types pooled
Duration and muscle-specific vulnerability as major determinants	Confirmed by tier 1 (variance components, coefficients) and by tier-2 importance

ignored in the primary model, and 547 rows do not state which leg was imaged.

6. **Mixed granularity and two estimators.** Composites and components coexist, resolved by a rule whose opposite (S1) has not been run; the target mixes the printed mean of individual changes with a value recomputed from group means.
7. **Publication bias** is plausible in this literature and cannot be quantified with 32 campaigns.
8. **Bed rest is an analogue.** Nothing here establishes that the coefficients transfer to microgravity; the 14 spaceflight rows are too few to test it and eight of them are lumbar multifidus.
9. **No data beyond 119 days**, and a single campaign at 119 days.
10. **Mostly young men.** 347 of the 425 lower-limb unloading rows come from men-only groups and 38 from older adults, in six campaigns.
11. **Group means, not individuals.** Every row is an arm mean; nothing can be said about individual susceptibility.
12. **The NASA campaign is thin.** Three of its four in-bed-rest occasions rest on one participant.
13. **Unsigned classification.** The muscle classification awaits physiological sign-off, a blocking condition for quoting coefficients by class.
14. **Assumptions of the model.** Campaigns are assumed independent of one another; missing dispersions and ages are assumed uninformative; the tier-2 families were fitted unweighted.
15. **Tier 3** is post hoc, overconfident, not bit-reproducible, and exposed to possible pretraining on the source papers.
16. **Most sensitivity analyses are outstanding**, including the three declared as must-show.

9 Status of the work and remaining tasks

The internal schedule placed the modelling hard stop on 21 September 2026; the report and slides follow, with the upload deadline on 10 October and the presentation on 15 or 16 October. The following items remain, in order of priority.

1. **Sensitivity analyses S1, S2 and S4** (composite-first resolution, without MEDES LTBR, MRI volume only), declared as must-show in this report; then the remaining analyses where they would change a conclusion.
2. **Physiological sign-off** of `data/muscle_map.csv` and of the fitted relationships by the scientific lead—a blocking criterion, and a line edit in one file.
3. **The 180-day extrapolation** with a prediction interval, as a clearly caveated backup result.
4. **Recalibration of tier 3's ranges** by a rule declared in advance, if its uncertainty is to be quoted.
5. **Reconcile the PRISMA counts** at the full-text stage with the final extraction.
6. **Close P3 and P4 formally**: tag the results and merge the branch stack ending at `feat/jev-figures` into `main`.
7. **Repository housekeeping before publication**: add a licence (code and dataset need different ones), remove the full-text digests that reproduce copyrighted text, acknowledge NASA as a data source, check the terms under which tier 3's cached answers may be published, and obtain both leads' agreement to the one schema change in v1.1.
8. **Two data checks**: two countermeasure-arm rows carry `cm_modality = none`, and the figure-derived extraction table fails the validator unless run in partial mode (23 rows lack sex or age).
9. **Author instructions** from the organisers (slide format, time limit, disclosure requirements), still outstanding.

10 Conclusions

The published bed-rest literature, organised into one frozen dataset of 742 comparisons from 52 studies and 36 campaigns, supports a quantified duration–response—about 3 percentage points of muscle per doubling of days of unloading, curved rather than linear—and a muscle-specific vulnerability led by the plantar flexors, with countermeasure arms losing measurably less. Tested honestly, by holding out whole campaigns, four machine-learning families add nothing to a simple duration curve, because the sample size is the number of campaigns, not the number of rows; a language model reading the other campaigns' data adds a small, robust but post-hoc improvement. The dataset and the framework are the contribution, and every number in this report can be regenerated from them with one command.

Data and code availability

The frozen dataset (`data/dataset_v1.1.csv`, tag `dataset-v1.1`), its card, the extraction schema, the campaign map, the muscle classification, the framework code, the configuration, every results file and the cached answers of the tier-3 model are held in the project repository [46]. `make all` regenerates every result from the dataset, and `report/make_figures.py` regenerates every table and figure in this report from the results. The source PDFs and the database exports are not redistributed, for copyright and licence reasons; the merge scripts and derived counts are included so that the search tables can be rebuilt from one's own exports. [\[Draft note: the repository has no licence file yet, and the full-text digests under `data/search/fulltext\_digests/` reproduce copyrighted text; both must be resolved before this statement can say](#)

the data are openly available ([docs/licensing.md](#)).]

Acknowledgements

Data for UTMB Campaign 3 were provided courtesy of the NASA Life Sciences Data Archive through the NASA Life Sciences Portal (experiment 4469ebdc-0a65-55e8-bbff-3cf6b044c4c6); we thank the principal investigators named on the experiment page. [Draft note: name the principal investigators from the NLSP experiment page, and add any institutional or funding acknowledgements.]

Author contributions

[Draft note: to be confirmed by both authors.] M. Bahari Qaragoz designed the extraction schema and the modelling framework, implemented the software, ran the analyses and drafted the report. [Co-author] designed and ran the literature search and screening, led the extraction and its source verification, and is responsible for the physiological interpretation. Both authors built the campaign map and reconciled the dataset.

Use of AI-based tools

The source verification of the extraction ([Section 3.7](#)) was AI-assisted. Tier 3 of the framework is itself a language model, used as a forecasting method under the controls described in [Section 4.6](#). [Draft note: state here, as required by the venue's policy, any AI assistance used in writing the code or this report.]

Competing interests and funding

[Draft note: to be completed.]

References

- [1] A. D. LeBlanc, V. S. Schneider, H. J. Evans, C. Pientok, R. Rowe, and E. Spector. Regional changes in muscle mass following 17 weeks of bed rest. *Journal of Applied Physiology*, 73(5):2172–2178, 1992. doi: 10.1152/jappl.1992.73.5.2172.
- [2] Björn A. Alkner and Per A. Tesch. Knee extensor and plantar flexor muscle size and function following 90 days of bed rest with or without resistance exercise. *European Journal of Applied Physiology*, 93(3):294–305, 2004. doi: 10.1007/s00421-004-1172-8.
- [3] Daniel L. Belavý, Tanja Miokovic, Gabriele Armbrrecht, Carolyn A. Richardson, Jörn Rittweger, and Dieter Felsenberg. Differential atrophy of the lower-limb musculature during prolonged bed-rest. *European Journal of Applied Physiology*, 107(4):489–499, 2009. doi: 10.1007/s00421-009-1136-0.
- [4] Tanja Miokovic, Gabriele Armbrrecht, Dieter Felsenberg, and Daniel L. Belavý. Heterogeneous atrophy occurs within individual lower limb muscles during 60 days of bed rest. *Journal of Applied Physiology*, 113(10):1545–1559, 2012. doi: 10.1152/japplphysiol.00611.2012.
- [5] T. A. Trappe, N. A. Burd, E. S. Louis, G. A. Lee, and S. W. Trappe. Influence of concurrent exercise or nutrition countermeasures on thigh and calf muscle size and function during 60 days of bed rest in women. *Acta Physiologica*, 191(2):147–159, 2007. doi: 10.1111/j.1748-1716.2007.01728.x.
- [6] Todd A. Trappe, Kiril Minchev, Ryan K. Perkins, Kaleen M. Lavin, Bozena Jemioło, Stephen M. Ratchford, Alex Claiborne, Gary A. Lee, W. Holmes Finch, Jeffrey W. Ryder, Lori Ploutz-Snyder, and Scott W. Trappe. NASA SPRINT exercise program efficacy for vastus lateralis and soleus skeletal muscle health during 70 days of simulated microgravity. *Journal of Applied Physiology*, 136(5):1015–1039, 2024. doi: 10.1152/japplphysiol.00489.2023.
- [7] Maude Dulac, Guy Hajj-Boutros, Vita Sonjak, Andréa Faust, Sabah N.A. Hussain, Stéphanie Chevalier, Isabelle J. Dionne, José A. Morais, and Gilles Gouspillou. A multimodal exercise countermeasure prevents the negative impact of head-down tilt bed rest on muscle volume and mitochondrial health in older adults. *The Journal of Physiology*, 603(13):3813–3836, 2024. doi: 10.1113/jp285897.
- [8] Daniel L Belavý, Hiroshi Ohshima, Jörn Rittweger, and Dieter Felsenberg. High-intensity flywheel exercise and recovery of atrophy after 90 days bed-rest. *BMJ Open Sport & Exercise Medicine*, 3(1):e000196, 2017. doi: 10.1136/bmjsem-2016-000196.
- [9] Uros Marusic, Marco Narici, Bostjan Simunic, Rado Pisot, and Ramona Ritzmann. Nonuniform loss of muscle strength and atrophy during bed rest: a systematic review. *Journal of Applied Physiology*, 131(1):194–206, 2021. doi: 10.1152/japplphysiol.00363.2020.
- [10] M. A. Johnson, J. Polgar, D. Weightman, and D. Appleton. Data on the distribution of fibre types in thirty-six human muscles: an autopsy study. *Journal of the Neurological Sciences*, 18(1):111–129, 1973. doi: 10.1016/0022-510X(73)90023-3.
- [11] Jochen Zange, Joachim Mester, Martina Heer, Götz Kluge, and Anna-Maria Liphardt. 20-Hz whole body vibration training fails to counteract the decrease in leg muscle volume caused by 14 days of 6° head down tilt bed rest. *European Journal of Applied Physiology*, 105(2):271–277, 2008. doi: 10.1007/s00421-008-0899-z.
- [12] Andreas Kramer, Albert Gollhofer, Gabriele Armbrrecht, Dieter Felsenberg, and Markus Gruber. How to prevent the detrimental effects of two months of bed-rest on muscle, bone and cardiovascular system: an RCT. *Scientific Reports*, 7(1), 2017. doi: 10.1038/s41598-017-13659-8.
- [13] Mirko Mandić, Tommy R. Lundberg, Jean-Pol Frippiat, Adam C. McDonnell, Igor B. Mekjavić, Marie-Pierre Bareille, Rebecca Billette de Villemeur, and Rodrigo Fernandez-Gonzalo. Limited musculoskeletal benefits of artificial gravity combined with cycling during bed rest: Results from the BRACE study. *Experimental Physiology*, 111(3):1181–1190, 2025. doi: 10.1113/ep093145.
- [14] Sofie K. Hansen, Pernille Hansen, Hanne Nygaard, Hans D. Grønbaek, Tania W. Berry, Camilla M. Olsen, Per Aagaard, Lars G. Hvid, Jakob Agergaard, Flemming Dela, and Charlotte Suetta. Five days of bed rest in young and old adults: Retainment of skeletal muscle mass with neuromuscular electrical stimulation. *Physiological Reports*, 12(16), 2024. doi: 10.14814/phy2.16166.
- [15] Cas J. Fuchs, Wesley J. H. Hermans, Jean Nyakayiru, Michelle E. G. Weijzen, Joey S. J. Smeets, Thorben Aussieker, Joan M. Senden, Will K. H. W. Wodzig, Tim Snijders, Lex B. Verdijk, and Luc J. C. van Loon.

- Daily blood flow restriction does not preserve muscle mass and strength during 2 weeks of bed rest. *The Journal of Physiology*, 603(13):3837–3856, 2024. doi: 10.1113/jp286065.
- [16] Mark Assink and Carlijn J. M. Wibbelink. Fitting three-level meta-analytic models in R: A step-by-step tutorial. *The Quantitative Methods for Psychology*, 12(3):154–174, 2016. doi: 10.20982/tqmp.12.3.p154.
- [17] Mathias Harrer, Pim Cuijpers, Toshi A. Furukawa, and David D. Ebert. *Doing Meta-Analysis with R: A Hands-On Guide*. Chapman and Hall/CRC, Boca Raton, FL, 2021. doi: 10.1201/9781003107347.
- [18] Wolfgang Viechtbauer. Conducting meta-analyses in R with the metafor package. *Journal of Statistical Software*, 36(3):1–48, 2010. doi: 10.18637/jss.v036.i03.
- [19] Alessio Crippa, Andrea Discacciati, Matteo Bottai, Donna Spiegelman, and Nicola Orsini. One-stage dose-response meta-analysis for aggregated data. *Statistical Methods in Medical Research*, 28(5):1579–1596, 2019. doi: 10.1177/0962280218773122.
- [20] Jonathan J. Deeks, Julian P. T. Higgins, and Douglas G. Altman. Analysing data and undertaking meta-analyses. In Julian P. T. Higgins and Sally Green, editors, *Cochrane Handbook for Systematic Reviews of Interventions*, chapter 9. The Cochrane Collaboration, version 5.1.0 edition, 2011. URL https://handbook-5-1.cochrane.org/chapter_9/9_6_4_meta_regression.htm. Section 9.6.4, Meta-regression.
- [21] Michael Geissbühler, Cesar A. Hincapié, Soheila Aghlmandi, Marcel Zwahlen, Peter Jüni, and Bruno R. da Costa. Most published meta-regression analyses based on aggregate data suffer from methodological pitfalls: a meta-epidemiological study. *BMC Medical Research Methodology*, 21:123, 2021. doi: 10.1186/s12874-021-01310-0.
- [22] Leo Breiman. Random forests. *Machine Learning*, 45(1):5–32, 2001. doi: 10.1023/A:1010933404324.
- [23] Jerome H. Friedman. Greedy function approximation: a gradient boosting machine. *The Annals of Statistics*, 29(5):1189–1232, 2001. doi: 10.1214/aos/1013203451.
- [24] Alex J. Smola and Bernhard Schölkopf. A tutorial on support vector regression. *Statistics and Computing*, 14(3):199–222, 2004. doi: 10.1023/B:STCO.0000035301.49549.88.
- [25] Tjeerd van der Ploeg, Peter C. Austin, and Ewout W. Steyerberg. Modern modelling techniques are data hungry: a simulation study for predicting dichotomous endpoints. *BMC Medical Research Methodology*, 14:137, 2014. doi: 10.1186/1471-2288-14-137.
- [26] David R. Roberts, Volker Bahn, Simone Ciuti, Mark S. Boyce, Jane Elith, Gurutzeta Guillera-Aroita, et al. Cross-validation strategies for data with temporal, spatial, hierarchical, or phylogenetic structure. *Ecography*, 40(8):913–929, 2017. doi: 10.1111/ecog.02881.
- [27] Scott M. Lundberg and Su-In Lee. A unified approach to interpreting model predictions. In *Advances in Neural Information Processing Systems 30 (NeurIPS 2017)*, pages 4765–4774, 2017.
- [28] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437.
- [29] Edward S. Epstein. A scoring system for probability forecasts of ranked categories. *Journal of Applied Meteorology*, 8(6):985–987, 1969. doi: 10.1175/1520-0450(1969)008<0985:ASSFPF>2.0.CO;2.
- [30] Todd A. Trappe, Per Tesch, Björn Alkner, and Scott Trappe. Microgravity-induced skeletal muscle atrophy in women and men: implications for long-duration spaceflights to the Moon and Mars. *Journal of Applied Physiology*, 135(5):1115–1119, 2023. doi: 10.1152/jappphysiol.00412.2023.
- [31] Jörn Rittweger, Klaus Möller, Marie-Pierre Bareille, Dieter Felsenberg, and Jochen Zange. Muscle X-ray attenuation is not decreased during experimental bed rest. *Muscle & Nerve*, 47(5):722–730, 2013. doi: 10.1002/mus.23644.
- [32] Kaitlyn R. Rogers, Gilhyeon Yoon, Caroline S. Vincenty, Scott W. Trappe, and Todd A. Trappe. Muscle-specific atrophy of the lower limb musculature in response to simulated microgravity exposure in women. *Journal of Applied Physiology*, 139(3):759–765, 2025. doi: 10.1152/jappphysiol.00483.2025.
- [33] Jacquelyn A. Holt, Brandon R. Macias, Robert Healey, Stuart M.C. Lee, Suzanne M.C. Schneider, Alexander J. Snyder, Douglas G. Chang, and Alan R. Hargens. WISE 2005: Aerobic and Resistive Exercises Protect Lumbar Paraspinal Lean Muscle Mass during 60 day Bed Rest in Women 2216 May 29, 10 30 AM - 10 45 AM. *Medicine & Science in Sports & Exercise*, 47(5S):586, 2015. doi: 10.1249/01.mss.0000478309.26945.3d.
- [34] Jacquelyn A. Holt, Brandon R. Macias, Suzanne M. Schneider, Donald E. Watenpaugh, Stuart M. C. Lee, Douglas G. Chang, and Alan R. Hargens. WISE 2005: Aerobic and resistive countermeasures prevent

- paraspinal muscle deconditioning during 60-day bed rest in women. *Journal of Applied Physiology*, 120(10): 1215–1222, 2016. doi: 10.1152/jappphysiol.00532.2015.
- [35] Guy Hajj-Boutros, Vita Sonjak, Andréa Faust, Eric Hedge, Carmelo Mastrandrea, Jean-Christophe Lagacé, Philippe St-Martin, Donya Naz Divsalar, Farshid Sadeghian, Stéphanie Chevalier, Teresa Liu-Ambrose, Andrew P. Blaber, Isabelle J. Dionne, Simon Duchesne, Richard Hughson, Saija Kontulainen, Olga Theou, and José A. Morais. Impact of 14 Days of Bed Rest in Older Adults and an Exercise Countermeasure on Body Composition, Muscle Strength, and Cardiovascular Function: Canadian Space Agency Standard Measures. *Gerontology*, 69(11):1284–1294, 2023. doi: 10.1159/000534063.
- [36] Jean-Christophe Lagacé, Philippe St-Martin, Mariano Avino, Ahmed Ghachem, Guy Hajj-Boutros, Jose A. Morais, Jamie S. McPhee, Karen Lambert, Eléonor Riesco, and Isabelle J. Dionne. Impact of 14 days of head-down bed rest and an exercise countermeasure on skeletal muscle atrophy, proteome and circulatory cytokines in older adults. *Experimental Physiology*, 2026. doi: 10.1113/ep093524.
- [37] NASA Life Sciences Data Archive. NASA Flight Analog Project, UTMB Campaign 3: whole-body DXA (experiment MR035G). NASA Life Sciences Portal (NLSP), experiment 4469ebdc-0a65-55e8-bbff-3cf6b044c4c6, 2026. URL <https://nlspace.nasa.gov/explore>. Open data, accessed 19 September 2026.
- [38] Kung-Yee Liang and Scott L. Zeger. Longitudinal data analysis using generalized linear models. *Biometrika*, 73(1):13–22, 1986. doi: 10.1093/biomet/73.1.13.
- [39] A. Colin Cameron and Douglas L. Miller. A practitioner’s guide to cluster-robust inference. *Journal of Human Resources*, 50(2):317–372, 2015. doi: 10.3368/jhr.50.2.317.
- [40] Frank E. Harrell, Jr. *Regression Modeling Strategies: With Applications to Linear Models, Logistic and Ordinal Regression, and Survival Analysis*. Springer Series in Statistics. Springer, Cham, 2nd edition, 2015. doi: 10.1007/978-3-319-19425-7.
- [41] Arthur E. Hoerl and Robert W. Kennard. Ridge regression: biased estimation for nonorthogonal problems. *Technometrics*, 12(1):55–67, 1970. doi: 10.1080/00401706.1970.10488634.
- [42] Fabian Pedregosa, Gaël Varoquaux, Alexandre Gramfort, Vincent Michel, Bertrand Thirion, Olivier Grisel, et al. Scikit-learn: machine learning in Python. *Journal of Machine Learning Research*, 12:2825–2830, 2011.
- [43] TypeSafe. Jev System One model, version jev-1.13.0. Hosted API, <https://api.typesafe.ai/v1/systemone>, 2026. Accessed 21 September 2026.
- [44] Marlou L. Dirks, Benjamin T. Wall, Bas van de Valk, Tanya M. Holloway, Graham P. Holloway, Adrian Chabowski, Gijs H. Goossens, and Luc J.C. van Loon. One Week of Bed Rest Leads to Substantial Muscle Atrophy and Induces Whole-Body Insulin Resistance in the Absence of Skeletal Muscle Lipid Accumulation. *Diabetes*, 65(10):2862–2875, 2016. doi: 10.2337/db15-1661.
- [45] Matthew J. Page, Joanne E. McKenzie, Patrick M. Bossuyt, Isabelle Boutron, Tammy C. Hoffmann, Cynthia D. Mulrow, et al. The PRISMA 2020 statement: an updated guideline for reporting systematic reviews. *BMJ*, 372:n71, 2021. doi: 10.1136/bmj.n71.
- [46] Milad Bahari Qaragoz and [Co-author]. From Bed Rest to Mars: dataset v1.1 and modelling framework. Git repository, tag dataset-v1.1, 2026. URL <https://github.com/MiladBahariQaragoz/BedrestToMars>.
- [47] Benjamin Lair, Elisa Le Roux, Isabelle de Glisezinski, Dominique Larrouy, Isabelle Harant, Marie-Pierre Bareille, Loïc Treffel, Guillemette Gauquelin-Koch, Cédric Moro, Chantal Simon, Claire Laurens, and Audrey Bergouignan. Five days of physical inactivity induced by dry immersion alter skeletal muscle metabolism and whole body glucose tolerance in healthy men. *Journal of Applied Physiology*, 140(1):279–293, 2026. doi: 10.1152/jappphysiol.00481.2025.
- [48] Ruth E. Tanner, Lucille B. Brunner, Jakob Agergaard, Katherine M. Barrows, Robert A. Briggs, Oh Sung Kwon, Laura M. Young, Paul N. Hopkins, Elena Volpi, Robin L. Marcus, Paul C. LaStayo, and Micah J. Drummond. Age-related differences in lean mass, protein synthesis and skeletal muscle markers of proteolysis after bed rest and exercise rehabilitation. *The Journal of Physiology*, 593(18):4259–4273, 2015. doi: 10.1113/jp270699.
- [49] E. Mulder, G. Clément, D. Linnarsson, W. H. Paloski, F. P. Wuyts, J. Zange, P. Frings-Meuthen, B. Johannes, V. Shushakov, M. Grunewald, N. Maassen, J. Buehlmeier, and J. Rittweger. Musculoskeletal effects of 5 days of bed rest with and without locomotion replacement training. *European Journal of Applied Physiology*, 115(4):727–738, 2014. doi: 10.1007/s00421-014-3045-0.
- [50] Benoit Smeuninx, Yasir S. Elhassan, Konstantinos N. Manolopoulos, Elizabeth Sapey, Alison B. Rushton, Sophie J. Edwards, Paul T. Morgan, Andrew Philp, Matthew S. Brook, Nima Gharahdaghi, Kenneth Smith,

- Philip J. Atherton, and Leigh Breen. The effect of short-term exercise prehabilitation on skeletal muscle protein synthesis and atrophy during bed rest in older men. *Journal of Cachexia, Sarcopenia and Muscle*, 12(1):52–69, 2020. doi: 10.1002/jcsm.12661.
- [51] Benoit Smeuninx, Yasir S. Elhassan, Elizabeth Sapey, Alison B. Rushton, Paul T. Morgan, Marie Korzepa, Archie E. Belfield, Andrew Philp, Matthew S. Brook, Nima Gharahdaghi, Daniel Wilkinson, Kenneth Smith, Philip J. Atherton, and Leigh Breen. A single bout of prior resistance exercise attenuates muscle atrophy and declines in myofibrillar protein synthesis during bed-rest in older men. *The Journal of Physiology*, 603(1): 87–105, 2023. doi: 10.1113/jp285130.
- [52] Micah J. Drummond, Kyle L. Timmerman, Melissa M. Markofski, Dillon K. Walker, Jared M. Dickinson, Mohammad Jamaluddin, Allan R. Brasier, Blake B. Rasmussen, and Elena Volpi. Short-term bed rest increases TLR4 and IL-6 expression in skeletal muscle of older adults. *American Journal of Physiology-Regulatory, Integrative and Comparative Physiology*, 305(3):R216–R223, 2013. doi: 10.1152/ajpregu.00072.2013.
- [53] Marlou L. Dirks, Joey S. J. Smeets, Andrew M. Holwerda, Imre W. K. Kouw, Gabriel N. Marzuca-Nassr, Annemie P. Gijzen, Graham P. Holloway, Lex B. Verdijk, and Luc J. C. van Loon. Dietary feeding pattern does not modulate the loss of muscle mass or the decline in metabolic health during short-term bed rest. *American Journal of Physiology-Endocrinology and Metabolism*, 316(3):E536–E545, 2019. doi: 10.1152/ajpendo.00378.2018.
- [54] Fabio Sarto, Daniel W. Stashuk, Martino V. Franchi, Elena Monti, Sandra Zampieri, Giacomo Valli, Giuseppe Sirago, Julián Candia, Lisa M. Hartnell, Matteo Paganini, Jamie S. McPhee, Giuseppe De Vito, Luigi Ferrucci, Carlo Reggiani, and Marco V. Narici. Effects of short-term unloading and active recovery on human motor unit properties, neuromuscular junction transmission and transcriptomic profile. *The Journal of Physiology*, 600(21):4731–4751, 2022. doi: 10.1113/jp283381.
- [55] Adam C. McDonnell, Ola Eiken, Petra Frings-Meuthen, Joern Rittweger, and Igor B. Mekjavic. The LunHab project: Muscle and bone alterations in male participants following a 10 day lunar habitat simulation. *Experimental Physiology*, 104(8):1250–1261, 2019. doi: 10.1113/ep087482.
- [56] Igor B. Mekjavic, Ola Eiken, Polona Jaki Mekjavic, and Adam C. McDonnell. Do females and males exhibit a similar sarcopenic response as a consequence of normoxic and hypoxic bed rest? *Experimental Physiology*, 106(1):37–51, 2020. doi: 10.1113/ep087834.
- [57] Martino V. Franchi, Fabio Sarto, Boštjan Šimunič, Rado Pišot, and Marco V. Narici. Early Changes of Hamstrings Morphology and Contractile Properties during 10 d of Complete Inactivity. *Medicine & Science in Sports & Exercise*, 54(8):1346–1354, 2022. doi: 10.1249/mss.0000000000002922.
- [58] Boštjan Šimunič, Fabio Sarto, Martino V. Franchi, Elena Monti, Rado Pišot, and Marco V. Narici. Alterations in Muscle Contractile Properties, Structure, and Function during 10-d Bed Rest, Postrecovery, and after COVID-19 Lockdown. *Medicine & Science in Sports & Exercise*, 58(8):1674–1686, 2026. doi: 10.1249/mss.0000000000003986.
- [59] Rado Pišot, Uros Marusic, Gianni Biolo, Sara Mazzucco, Stefano Lazzer, Bruno Grassi, Carlo Reggiani, Luana Toniolo, Pietro Enrico di Prampero, Angelina Passaro, Marco Narici, Shahid Mohammed, Joern Rittweger, Mladen Gasparini, Mojca Gabrijelčič Blenkuš, and Boštjan Šimunič. Greater loss in muscle mass and function but smaller metabolic alterations in older compared with younger men following 2 wk of bed rest and recovery. *Journal of Applied Physiology*, 120(8):922–929, 2016. doi: 10.1152/japplphysiol.00858.2015.
- [60] Enrico Rejc, Mirco Floreani, Paolo Taboga, Alberto Botter, Luana Toniolo, Lina Cancellara, Marco Narici, Boštjan Šimunič, Rado Pišot, Gianni Biolo, Angelina Passaro, Joern Rittweger, Carlo Reggiani, and Stefano Lazzer. Loss of maximal explosive power of lower limbs after 2 weeks of disuse and incomplete recovery after retraining in older adults. *The Journal of Physiology*, 596(4):647–665, 2018. doi: 10.1113/jp274772.
- [61] Cas J. Fuchs, Wesley J. H. Hermans, Job van den Hurk, Christopher J. Wiggins, Per Widholm, Olof Dahlqvist Leinhard, Pandichelvam Veeraiah, Joachim E. Wildberger, Jeanine J. Prompers, and Luc J. C. van Loon. Quantifying Leg Muscle Disuse Atrophy During Bed Rest Using DXA, CT, and MRI. *European Journal of Sport Science*, 25(5), 2025. doi: 10.1002/ejsc.12299.
- [62] Mira A. Orlova, Nadia S. Kurochkina, Roman Yu. Zhedyaev, Nikita E. Vavilov, Viktor G. Zgoda, Alina A. Puchkova, Anna A. Borzykh, Tatiana F. Vepkhvadze, Egor M. Lednev, Pavel A. Makhnovskii, Gleb V. Galkin, Valery E. Novikov, Alexey V. Shpakov, and Daniil V. Popov. Different effects of 3-week disuse on phenotype and gene expression in calf and thigh muscles. *The Journal of Physiology*, 604(12):4851–4875, 2026. doi: 10.1113/jp290722.

- [63] Tadej Debevec, Bergita Ganse, Uwe Mittag, Ola Eiken, Igor B. Mekjavic, and Jörn Rittweger. Hypoxia Aggravates Inactivity-Related Muscle Wasting. *Frontiers in Physiology*, 9, 2018. doi: 10.3389/fphys.2018.00494.
- [64] Anna-Maria Liphardt, Vera Bolte, Felix Eckstein, Wolfgang Wirth, Gert-Peter Brüggemann, and Anja Niehoff. Response of thigh muscle cross-sectional area to 21-days of bed rest with exercise and nutrition countermeasures. *Translational Sports Medicine*, 3(2):93–106, 2019. doi: 10.1002/tsm2.122.
- [65] John E. Greenleaf, P. L. Lee, S. Ellis, R. H. Selzer, and D. A. Ortendahl. Leg muscle volume during 30-day 6-degree head-down bed rest with isotonic and isokinetic exercise training. NASA Technical Memorandum 4580, NASA Ames Research Center, 1994. NTRS accession N94-29401.
- [66] Summer B. Cook, Jill A. Kanaley, and Lori L. Ploutz-Snyder. Neuromuscular function following muscular unloading and blood flow restricted exercise. *European Journal of Applied Physiology*, 114(7):1357–1365, 2014. doi: 10.1007/s00421-014-2864-3.
- [67] Hans E. Berg, Ola Eiken, Lucijan Miklavcic, and Igor B. Mekjavic. Hip, thigh and calf muscle atrophy and bone loss after 5-week bedrest inactivity. *European Journal of Applied Physiology*, 99(3):283–289, 2006. doi: 10.1007/s00421-006-0346-y.
- [68] Felix Krainski, Jeffrey L. Hastings, Katja Heinicke, Nadine Romain, Eric L. Pacini, Peter G. Snell, Phil Wyrick, M. Dean Palmer, Ronald G. Haller, and Benjamin D. Levine. The effect of rowing ergometry and resistive exercise on skeletal muscle structure and function during bed rest. *Journal of Applied Physiology*, 116(12):1569–1581, 2014. doi: 10.1152/japplphysiol.00803.2013.
- [69] P. Arbeille, K. Zuj, and L. Guillon. Exercise combined with artificial gravity and exercise only countermeasures prevent organ and blood vessel morphological changes induced by 55 days HDT bedrest. *Frontiers in Physiology*, 15, 2024. doi: 10.3389/fphys.2024.1482860.
- [70] Enrico De Martino, Julie Hides, James M. Elliott, Mark Hoggarth, Jochen Zange, Kirsty Lindsay, Dorothée Debusse, Andrew Winnard, David Beard, Jonathan A. Cook, Sauro E. Salomoni, Tobias Weber, Jonathan Scott, Paul W. Hodges, and Nick Caplan. Lumbar muscle atrophy and increased relative intramuscular lipid concentration are not mitigated by daily artificial gravity after 60-day head-down tilt bed rest. *Journal of Applied Physiology*, 131(1):356–368, 2021. doi: 10.1152/japplphysiol.00990.2020.
- [71] Enrico De Martino, Julie Hides, James M. Elliott, Mark A. Hoggarth, Jochen Zange, Kirsty Lindsay, Dorothée Debusse, Andrew Winnard, David Beard, Jonathan A. Cook, Sauro E. Salomoni, Tobias Weber, Jonathan Scott, Paul W. Hodges, and Nick Caplan. The Effects of Reconditioning Exercises Following Prolonged Bed Rest on Lumbopelvic Muscle Volume and Accumulation of Paraspinal Muscle Fat. *Frontiers in Physiology*, 13, 2022. doi: 10.3389/fphys.2022.862793.
- [72] Vienna Tran, Enrico De Martino, Julie Hides, Gordon Cable, James M. Elliott, Mark Hoggarth, Jochen Zange, Kirsty Lindsay, Dorothée Debusse, Andrew Winnard, David Beard, Jonathan A. Cook, Sauro E. Salomoni, Tobias Weber, Jonathan Scott, Paul W. Hodges, and Nick Caplan. Gluteal Muscle Atrophy and Increased Intramuscular Lipid Concentration Are Not Mitigated by Daily Artificial Gravity Following 60-Day Head-Down Tilt Bed Rest. *Frontiers in Physiology*, 12, 2021. doi: 10.3389/fphys.2021.745811.
- [73] Peter R. Cavanagh, Andrea J. Rice, Sara C. Novotny, Kerim O. Genc, Ricki K. Englehaupt, Tammy M. Owings, Bryan Comstock, Tamre Cardoso, Hakan Ilaslan, Scott M. Smith, and Angelo A. Licata. Replacement of daily load attenuates but does not prevent changes to the musculoskeletal system during bed rest. *Bone Reports*, 5: 299–307, 2016. doi: 10.1016/j.bonr.2016.10.001.
- [74] J. A. Hides, G. Lambrecht, W. R. Stanton, and V. Damann. Changes in multifidus and abdominal muscle size in response to microgravity: possible implications for low back pain research. *European Spine Journal*, 25 (S1):175–182, 2015. doi: 10.1007/s00586-015-4311-5.
- [75] Julie A. Hides, Gunda Lambrecht, Christopher T. Sexton, Casey Pruett, Nora Petersen, Patrick Jaekel, André Rosenberger, and Guillaume Weerts. The effects of exposure to microgravity and reconditioning of the lumbar multifidus and anterolateral abdominal muscles: implications for people with LBP. *The Spine Journal*, 21(3): 477–491, 2021. doi: 10.1016/j.spinee.2020.09.006.
- [76] Jonas Böcker, Patrick Lau, Uwe Mittag, Wilhelm Bloch, Laurence Vico, and Jörn Rittweger. Comparison of musculoskeletal responses and its variability after long-term spaceflight and prolonged bed rest conditions. *npj Microgravity*, 12(1), 2026. doi: 10.1038/s41526-026-00611-2.

A Search strategies

All searches were run on 4 September 2026 with a publication-date limit of 2013 onwards applied in each interface. The strings below are reproduced verbatim from docs/literature-review/search_log.md.

PubMed. (1,412 records; filters: humans, 2013 onwards)

```
("Bed Rest"[Mesh] OR "Head-Down Tilt"[Mesh] OR "Weightlessness Simulation"[Mesh]
OR "Immobilization"[Mesh] OR "bed rest"[tiab] OR bedrest[tiab] OR "bed-rest"[tiab]
OR "head-down tilt"[tiab] OR "head down tilt"[tiab] OR "head-down bed rest"[tiab]
OR HDBR[tiab] OR "6 degrees head-down"[tiab] OR antiorthostatic[tiab]
OR "anti-orthostatic"[tiab] OR hypokinesia[tiab] OR hypodynamia[tiab]
OR "dry immersion"[tiab] OR "unilateral lower limb suspension"[tiab] OR ULLS[tiab]
OR "limb suspension"[tiab] OR "simulated microgravity"[tiab]
OR "microgravity analogue"[tiab] OR "microgravity analog"[tiab]
OR "spaceflight analogue"[tiab] OR "spaceflight analog"[tiab]
OR "ground-based analogue"[tiab] OR "mechanical unloading"[tiab]
OR "muscle unloading"[tiab] OR disuse[tiab])
AND
("Muscular Atrophy"[Mesh] OR "Muscle, Skeletal"[Mesh] OR atroph*[tiab]
OR "muscle volume"[tiab] OR "muscle mass"[tiab] OR "muscle size"[tiab]
OR "cross-sectional area"[tiab] OR CSA[tiab] OR PCSA[tiab] OR "lean mass"[tiab]
OR "lean tissue"[tiab] OR "muscle thickness"[tiab] OR "muscle wasting"[tiab]
OR deconditioning[tiab] OR soleus[tiab] OR gastrocnemius[tiab]
OR "triceps surae"[tiab] OR "plantar flexor"[tiab] OR quadriceps[tiab]
OR "vastus lateralis"[tiab] OR "knee extensor"[tiab])
AND humans[Filter]
```

Scopus. (1,757 records)

```
TITLE-ABS-KEY(
("bed rest" OR bedrest OR "head-down tilt" OR "head down bed rest" OR HDBR
OR antiorthostatic OR hypokinesia OR hypodynamia OR "dry immersion"
OR "limb suspension" OR ULLS OR "simulated microgravity" OR "microgravity analog*"
OR "spaceflight analog*" OR "mechanical unloading" OR "muscle unloading" OR disuse)
AND
(atroph* OR "muscle volume" OR "muscle mass" OR "muscle size"
OR "cross-sectional area" OR PCSA OR "lean mass" OR "muscle thickness"
OR "muscle wasting" OR deconditioning OR soleus OR gastrocnemius
OR "triceps surae" OR quadriceps OR "vastus lateralis" OR "knee extensor")
)
AND NOT TITLE-ABS-KEY(rodent OR mice OR mouse OR rat OR hindlimb OR "hind limb")
AND DOCTYPE(ar OR cp OR re)
```

Web of Science Core Collection. (1,876 records)

```
TS=(("bed rest" OR bedrest OR "head-down tilt" OR "head down bed rest" OR HDBR
OR antiorthostatic OR hypokinesia OR hypodynamia OR "dry immersion"
OR "limb suspension" OR ULLS OR "simulated microgravity"
OR "microgravity analog*" OR "spaceflight analog*" OR "mechanical unloading"
OR "muscle unloading" OR disuse)
AND
(atroph* OR "muscle volume" OR "muscle mass" OR "muscle size"
OR "cross-sectional area" OR PCSA OR "lean mass" OR "muscle thickness"
OR "muscle wasting" OR deconditioning OR soleus OR gastrocnemius
OR "triceps surae" OR quadriceps OR "vastus lateralis" OR "knee extensor"))
NOT TS=(rat OR rats OR mice OR mouse OR rodent OR hindlimb OR "hind limb")
```

NASA Technical Reports Server. (686 records; five queries)

"bed rest" "muscle volume"
 "bed rest" muscle atrophy
 "head-down tilt" muscle
 "antiorthostatic" muscle
 bed rest countermeasure exercise muscle

B Extraction schema

Table 23 lists the 61 columns of the frozen schema by group. The full definitions, units, controlled vocabularies and validation checks are in `data/schema.md`.

Table 23. The 61 columns of the frozen extraction schema. One row is one study, arm, muscle and measurement occasion, compared with that arm's baseline.

Group	Columns
Identity and provenance	row_id, study_id, cohort_id, campaign_name, registry_id, first_author, year, doi, source_file
Unloading design	design, hdt_angle_deg, duration_days, phase, timepoint_days, days_from_unloading_end, exposure_flag
Arm and participants	arm_id, arm_type, cm_modality, cm_dose, n_arm, n_analysed, sex, pct_female, age_mean, age_sd, age_min, age_max, population, bmi_mean, body_mass_mean_kg, nutrition_controlled
What was measured	muscle, muscle_function_class, is_composite, composite_of, laterality, measurement_site, outcome_type, modality
Values	unit_original, value_baseline_original, value_followup_original, unit_si, value_baseline, value_followup, change_absolute, pct_change, variance_of, variance_type, variance_value, p_value
Provenance and quality control	data_source, digitizer_tool, page_ref, extractor, extraction_date, extraction_confidence, double_extracted, qc_flag, notes

C The 36 campaigns

Table 24. The 36 independent campaigns of dataset_v1.1, ordered by planned length. Design: HDBR, head-down tilt bed rest; HBR, horizontal bed rest; ULLS, unilateral lower-limb suspension; DI, dry immersion; SF, spaceflight. Studies: the number of papers reporting the campaign in the dataset, with their references. Rows: all rows, and rows in modelling subset A. Registry identifiers are shown where known.

Campaign	Design	Days	Studies	Modality	Rows	A
Toulouse dry immersion NCT03915457	DI	5	1 [47]	MRI	1	1
Utah BR-5	HBR	5	1 [48]	DXA	2	2
DLR HDT-5 crossover	HDBR	5	1 [49]	MRI	6	6
Copenhagen BR-5	HBR	5	1 [14]	DXA, ultrasound	12	12
Birmingham BR-5 NCT04422665	HBR	5	2 [50, 51]	MRI	24	8
BR-7, older adults (Drummond)	HBR	7	1 [52]	DXA	1	1
Maastricht BR-7	HBR	7	1 [44]	CT	1	1
Maastricht BR-7, feeding	HBR	7	1 [53]	CT	2	2
ULLS-10 (Sarto)	ULLS	10	1 [54]	ultrasound	1	1
LunHab	HBR	10	1 [55]	DXA	4	4
PlanHab BR-10	HBR	10	1 [56]	CT	4	4
BR-10 (Franchi, Šimunič)	HBR	10	2 [57, 58]	MRI, ultrasound	9	8
Izola BR-14	HBR	14	2 [59, 60]	MRI	3	3
Maastricht BR-14	HBR	14	2 [15, 61]	CT, DXA, MRI	7	7
McGill HDBR-14 NCT04964999	HDBR	14	3 [7, 35, 36]	DXA, MRI	8	4
DLR VBR	HDBR	14	1 [11]	MRI	20	8
HDBR-21 (Orlova)	HDBR	21	1 [62]	DXA	2	2
PlanHab BR-21	HBR	21	1 [63]	CT	2	2
HDBR-21 (Liphardt)	HDBR	21	1 [64]	MRI	20	20
NASA Ames HDBR-30	HDBR	30	1 [65]	MRI	3	0
ULLS-30 (Cook)	ULLS	30	1 [66]	MRI	4	4
Valdoltra BR-35	HBR	35	1 [67]	CT	6	3
Dallas HDBR-35	HDBR	35	1 [68]	MRI	8	8
Berlin BBR1	HBR	56	1 [3]	MRI	68	52
DLR RSL DRKS00012946	HDBR	60	1 [12]	DXA	2	0
BRACE	HDBR	60	2 [13, 69]	MRI, ultrasound	18	18
WISE-2005	HDBR	60	4 [5, 32–34]	MRI	35	26
Berlin BBR2-2	HDBR	60	1 [4]	MRI	38	30
AGBRESA DRKS00015677	HDBR	60	3 [70–72]	MRI	87	9
NASA SPRINT	HDBR	70	1 [6]	MRI	9	3
HDBR-84 (Cavanagh)	HDBR	84	1 [73]	MRI	2	2
NASA UTMB Campaign 3	HDBR	90	1 [37]	DXA	5	4
MEDES LTBR	HDBR	90	4 [2, 8, 30, 31]	CT, MRI	299	86
NASA 17-week BR	HBR	119	1 [1]	DXA, MRI	9	5

Table 24, continued

Campaign	Design	Days	Studies	Modality	Rows	A
ISS astronauts (Hides)	SF	180	2 [74 , 75]	ultrasound	8	0
Spaceflight vs. BR (Böcker)	SF	180	1 [76]	CT	12	0

D Muscle classification

Table 25. Muscle classification from data/muscle_map.csv: family, functional class, fibre-type profile and the written rationale for each of the 51 muscles and muscle groups. Joined to the dataset at load time. Awaiting physiological sign-off.

Muscle	Family	Class	Fibre profile	Rationale
anterior tibial group	Dorsiflexors	Flexor	slow dominant	Ankle dorsiflexors; they lift the foot rather than support the body
extensor digitorum longus	Dorsiflexors	Flexor	unknown	Extends the toes but dorsiflexes the ankle; the name refers to the toes not the ankle
tibialis anterior	Dorsiflexors	Flexor	slow dominant	Ankle dorsiflexor. Slow-fibre dominant but not weight-bearing - the case that separates fibre type from mechanical role
peroneals	Evertors	Mixed	unknown	Evert the foot and assist plantarflexion weakly; a stabiliser rather than a supporter
gluteus medius	Hip abductors	Mixed	unknown	Frontal-plane pelvic stabiliser rather than an extensor
gluteus minimus	Hip abductors	Mixed	unknown	Frontal-plane pelvic stabiliser rather than an extensor
adductor brevis	Hip adductors	Mixed	unknown	Hip adduction with a minor flexion component; no role in supporting body weight against gravity
adductor longus	Hip adductors	Mixed	unknown	Hip adduction with a minor flexion component; not an antigravity extensor
adductor magnus	Hip adductors	Mixed	unknown	Adductor with a posterior head that extends the hip; part extensor part adductor
gracilis	Hip adductors	Mixed	unknown	Adducts the hip and assists knee flexion; crosses two joints in neither an extensor role
gluteals	Hip extensors	Mixed	unknown	Reported as one block; mixes the hip extensor with the two abductors
gluteus maximus	Hip extensors	Antigravity	unknown	Principal hip extensor; resists trunk flexion against gravity in stance and stair climbing
iliopsoas	Hip flexors	Flexor	unknown	Principal hip flexor; unloaded rather than loaded by upright stance
pectineus	Hip flexors	Flexor	unknown	Flexes and adducts the hip
psoas	Hip flexors	Flexor	unknown	Hip flexor; unloaded rather than loaded by upright stance
sartorius	Hip flexors	Flexor	unknown	Flexes and externally rotates the hip and assists knee flexion
obturator externus	Hip rotators	Mixed	unknown	Deep external rotator and joint stabiliser
obturator internus	Hip rotators	Mixed	unknown	Deep external rotator and joint stabiliser
quadratus femoris	Hip rotators	Mixed	unknown	Deep external rotator and joint stabiliser
anterior thigh compartment	Knee extensors	Antigravity	mixed	Compartment measurement dominated by the quadriceps
quadriceps	Knee extensors	Antigravity	mixed	Principal knee extensor group; resists knee collapse under body weight
rectus femoris	Knee extensors	Mixed	mixed	Crosses two joints - extends the knee but flexes the hip; loses less than the vasti in bed rest
vasti	Knee extensors	Antigravity	mixed	The three monoarticular knee extensors reported as one value
vastus intermedius	Knee extensors	Antigravity	mixed	Monoarticular knee extensor
vastus lateralis	Knee extensors	Antigravity	fast leaning	Monoarticular knee extensor
vastus medialis	Knee extensors	Antigravity	mixed	Monoarticular knee extensor

Table 25, continued

Muscle	Family	Class	Fibre profile	Rationale
biceps femoris long head	Knee flexors	Mixed	unknown	Crosses two joints - extends the hip and flexes the knee
biceps femoris short head	Knee flexors	Flexor	unknown	Crosses the knee only and flexes it; no hip extension role
hamstrings	Knee flexors	Mixed	unknown	Group value; the biarticular members extend the hip while all of them flex the knee
lateral hamstrings	Knee flexors	Mixed	unknown	Group value for the two biceps femoris heads
medial hamstrings	Knee flexors	Mixed	unknown	Group value for the two medial hamstrings
popliteus	Knee flexors	Mixed	unknown	Unlocks and rotates the knee; a small stabiliser
posterior thigh compartment	Knee flexors	Mixed	unknown	Compartment measurement dominated by the hamstrings
semimembranosus	Knee flexors	Mixed	unknown	Crosses two joints - extends the hip and flexes the knee
semitendinosus	Knee flexors	Mixed	unknown	Crosses two joints - extends the hip and flexes the knee
flexor digitorum longus	Plantar flexors	Antigravity	unknown	Flexes the toes but plantarflexes the ankle; contributes to push-off and postural support
flexor digitorum with tibialis posterior	Plantar flexors	Antigravity	unknown	Deep posterior compartment measured as one block; both members plantarflex the ankle
flexor hallucis longus	Plantar flexors	Antigravity	unknown	Flexes the great toe but plantarflexes the ankle
gastrocnemius lateralis	Plantar flexors	Antigravity	mixed	Biarticular plantarflexor loaded during standing and walking
gastrocnemius medialis	Plantar flexors	Antigravity	mixed	Biarticular plantarflexor loaded during standing and walking
gastrocnemius total	Plantar flexors	Antigravity	mixed	Both gastrocnemius heads reported as one value
plantar flexors	Plantar flexors	Antigravity	mixed	Whole posterior compartment reported as one value
soleus	Plantar flexors	Antigravity	slow dominant	Monoarticular plantarflexor carrying continuous postural load in stance; roughly 80% type I fibres
tibialis posterior	Plantar flexors	Antigravity	unknown	Plantarflexes and inverts the ankle and supports the medial arch
triceps surae	Plantar flexors	Antigravity	mixed	Soleus plus both gastrocnemius heads reported as one value
lumbar erector spinae	Trunk extensors	Antigravity	unknown	Spinal extensor that holds the trunk upright; antigravity but not lower limb
multifidus	Trunk extensors	Antigravity	slow dominant	Deep segmental spinal extensor and postural stabiliser; antigravity but not lower limb
quadratus lumborum	Trunk extensors	Mixed	unknown	Lateral flexor and stabiliser of the lumbar spine rather than a pure extensor
whole calf	Whole segment	Mixed	mixed	Whole-segment measurement spanning muscles of every functional class
whole lower limb	Whole segment	Mixed	mixed	Whole-segment measurement spanning muscles of every functional class
whole thigh	Whole segment	Mixed	mixed	Whole-segment measurement spanning muscles of every functional class

E Tier-1 coefficients

Table 26. Tier-1 fixed effects, in percentage points, with 95% cluster-robust confidence intervals: the saturating form on subset A (346 rows, 32 campaigns, $\tau = 60$ days, t_{31}) and the ranking model on subset B (304 rows, 25 campaigns, $\tau = 60$ days, t_{24}). The duration coefficient multiplies $1 - e^{-t/\tau}$ and so equals the eventual change relative to the intercept. Subset B has no whole-segment rows and no DXA rows.

Term	Subset A, saturating		Subset B, ranking model	
	Estimate (95% CI)	<i>p</i>	Estimate (95% CI)	<i>p</i>
Intercept	−3.04 (−6.20 to 0.13)	0.059	−1.26 (−4.11 to 1.60)	0.373
$1 - e^{-t/\tau}$	−14.25 (−16.40 to −12.11)	<0.001	−13.98 (−16.84 to −11.12)	<0.001
<i>Muscle family (vs. knee extensors)</i>				
Dorsiflexors	−0.32 (−3.03 to 2.39)	0.813	−0.08 (−2.61 to 2.44)	0.945
Hip abductors	0.95 (−1.87 to 3.77)	0.497	1.01 (−1.65 to 3.68)	0.440
Hip adductors	4.72 (2.19 to 7.25)	<0.001	4.70 (2.12 to 7.28)	<0.001
Hip extensors	4.33 (0.83 to 7.82)	0.017	4.03 (0.79 to 7.27)	0.017
Hip flexors	3.33 (−0.03 to 6.69)	0.052	3.33 (−0.13 to 6.80)	0.059
Hip rotators	8.37 (5.98 to 10.75)	<0.001	8.22 (5.89 to 10.56)	<0.001
Knee flexors	0.23 (−2.31 to 2.76)	0.857	0.28 (−2.36 to 2.92)	0.829
Plantar flexors	−5.69 (−8.91 to −2.48)	0.001	−5.69 (−8.95 to −2.42)	0.001
Whole segment	−2.54 (−7.05 to 1.97)	0.260	—	—
Countermeasure arm	3.43 (1.68 to 5.18)	<0.001	3.73 (1.54 to 5.93)	0.002
<i>Modality and outcome (vs. CT CSA)</i>				
DXA lean mass	4.75 (0.05 to 9.45)	0.048	—	—
MRI CSA	0.73 (−3.64 to 5.09)	0.736	−1.07 (−5.01 to 2.87)	0.581
MRI volume	0.49 (−2.25 to 3.23)	0.718	−1.71 (−3.54 to 0.12)	0.066
Ultrasound CSA	1.11 (−1.27 to 3.49)	0.349	−0.52 (−1.99 to 0.94)	0.468
Ultrasound thickness	5.95 (0.61 to 11.29)	0.030	3.88 (−0.52 to 8.28)	0.081
Composite muscle	−0.38 (−2.73 to 1.96)	0.740	−0.57 (−2.79 to 1.64)	0.598

F Tier-3 recognition probe by campaign

Table 27. Recognition probe: for each of the 11 campaigns with a proper name, the rows probed, the mean probability the model placed on the true name, the share of rows on which the true name was its first choice, its commonest first choice, and the campaign’s paired MAE gain over the curve in the forecast without history.

Campaign	Rows	Mean $p(\text{true})$	Named first	Commonest first choice	Gain (pp)
NASA SPRINT	3	0.55	67%	NASA SPRINT	+2.95
Berlin BBR2-2	30	0.04	0%	None of these	+2.80
MEDES LTBR	86	0.18	41%	None of these	+1.75
WISE-2005	26	0.04	0%	MEDES Long Term Bed Rest (LTBR)	+1.50
Berlin BBR1	52	0.11	0%	None of these	+0.86
BRACE	18	0.06	0%	None of these	+0.80
NASA UTMB Campaign 3	4	0.06	0%	None of these	+0.67
DLR VBR	8	0.37	50%	Vibration Bed Rest Study (VBR)	−0.04
AGBRESA	9	0.45	67%	AGBRESA	−0.27
PlanHab BR-21	2	0.03	0%	None of these	−0.46
LunHab	4	0.04	0%	None of these	−0.48

G Reproducing the results

From a clean checkout of the repository at the commit named on the title page:

```
make venv # creates a virtual environment from requirements.txt
PYTHON=~/.venvs/dglrm/bin/python make test # 223 checks in 19 files
PYTHON=~/.venvs/dglrm/bin/python make all # tier 1, S6, baseline, the four families
make forecast ablation validation # tier 3, rebuilt from results/forecast_cache/
make figures # slide figures F1-F9
python report/make_figures.py # this report's figures and tables
cd report && pdflatex main && bibtex main && pdflatex main && pdflatex main
```

The loader refuses to run if `data/dataset_v1.1.csv` no longer matches its recorded SHA-256 (`dd214c6a a33dee7ab62358bc56c1e6a72eb330085484022ba1ee2bad394395b9`), so no result can come from an edited dataset. `make forecast-live` asks the TypeSafe service for any answer the cache lacks and needs an API key; because the service’s answers vary slightly between identical requests, a live rebuild moves individual forecasts by about 0.17 pp ([Section 6.8](#)).